Software Engineering Document (SED)

Project: Settle

Client: Zillow

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- Header Page
- Table of Contents
- Introduction to Document
- Requirements / System Modeling
- Wire Frames
- Software Testing
- Glossary of Terms

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- Architecture Design

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1. Introduction of Document

1.1 Purpose of this Document

The Software Engineering Document describes the design, architecture, functional structure and system design for Settle web applications. Settle is an apartment recommendation platform that helps relocate renters to find personalized housing options based on their preferences, lifestyle priorities, and interaction history.

1.2 Audience

This document is intended for project managers, software engineers, and other stakeholders involved in the design, development, and implementation of the Settle System.

1.3 Structure

This document outlines the design and implementation details of Settle. The system is composed of several major components, including the User Interface, Recommendation Engine, Listing and Data Service, Personalized System, Search and Filtering System, Backend Application Services, and Database. Each component is described in detail in the following sections.

2. Requirements / System Modeling

2.1 Approach to Gather Requirements

The Settle team used an iterative and documentation-driven approach to gather requirements. Initial requirements were gathered through stakeholder interviews, which analyzed user workflows and focused on how they would interact with the various web pages offered in the web application. Each stakeholder role was chosen to get a deep understanding of the expectations for results and system expectations. All requirements were formally documented in a shared Software Engineering Document, allowing the team to track updates and maintain consistency throughout the project lifecycle.

In addition to gathering requirements, the team conducted weekly review sessions to assess documented requirements, identify gaps, and clarify ambiguities. Feedback obtained from these review sessions was used to improve the wording of requirements, validate assumptions, and ensure consistency in all parts. This structured approach enabled the team to enhance shared understanding of system behavior and helped produce a well-defined initial draft of the software engineering document.

2.2 Functional Requirements (FR)

Rithanya Dhivakar

FR-01 The system shall collect user preferences using structured inputs, including sliders, checkboxes, and selection controls, to capture constraints such as budget, commute tolerance, safety priority, amenities, and renter experience level.

FR-02 The system shall generate and display recommendation results that reflect the user's stated preferences and strongest constraints, including budget range, commute tolerance, and safety priority, and present each recommendation as a listing card showing key details such as price, location, commute, safety rating, and a match indicator.

FR-03 The system shall adapt the presentation of recommendation results based on the user's experience level, providing additional guidance and explanations for first-time renters and more comparative information for experienced renters.

FR-04 The system shall display plain-language explanations for each recommendation that describe the primary factors influencing the result and any notable trade-offs and shall present these explanations as concise bullet-style highlights summarizing the primary influencing factors.

FR-05 The system shall allow users to view additional details behind a recommendation, such as neighborhood context or listing information, without requiring them to view raw or complex data.

FR-06 The system shall display a summary of the user's current preferences on the results page, including budget range, commute limit, and safety priority.

FR-07 The system shall track user interactions with recommendations, including viewed listings and saved listings, and use this interaction history to adjust future recommendation results while respecting stated preferences, and shall allow users to access recently viewed listings from their interaction history.

FR-08 The system shall display feedback indicating how user interactions, such as saved listings or repeated exploration of similar neighborhoods, have influenced current recommendations, including a summary of the main factors influencing the recommendation focus.
FR-09 The system shall allow users to refine or change recommendation focus through a free-text input and shall interpret the input to update relevant preferences and regenerate recommendation results.
FR-10 The system shall notify users when no strong recommendations match their current preferences and shall suggest actionable alternatives, such as adjusting constraints or exploring nearby neighborhoods.

Jose Rogel
FR-11 The system must allow users to search for apartments by zip code, neighborhood name, or keyword and return a filtered list of matching listings.
FR-12 The system must allow users to filter apartment search results by total monthly cost, number of bedrooms, pet policy, and proximity to workplace.
FR-13 The system must display a detailed apartment listing page showing rent, address, available amenities, neighborhood safety score, a direct messaging option, and secondary contact information for the property.
FR-14 The system must display nearby amenities for a selected apartment, including grocery stores, gyms, parks, restaurants, coworking spaces, and entertainment venues, along with their distance from the listing.
FR-15 The system must allow users to save apartment listings to a persistent favorites list accessible across devices from their profile.
FR-16 The system must display an About page describing the platform's mission, the team behind it, and the data sources used to generate safety and compatibility insights.
FR-17 The system must provide a Contact Us page with a form that allows users to submit inquiries, which are then routed to the platform support team via email.
FR-18 The system must display a Privacy Policy page outlining how user data is collected, stored, and used.
FR-19 The system must display a Terms of Service page that users must acknowledge before creating an account.
FR-20 The system must include a FAQ page accessible from the footer that addresses common questions about compatibility scoring, data sources, and account management.

Shreya Sil
FR-21: The system shall validate all user preference inputs (budget range, commute time, amenities) to ensure logical consistency before generating recommendations.
FR-22: The system shall integrate with Zillow's live rental listing database to retrieve up-to-date property information in real time.
FR-23: The system shall rank recommended listings based on weighted user preferences and display them in prioritized order.
FR-24: The system shall allow users to save, compare, and remove listings from a personalized shortlist.

FR-25: The system shall automatically refresh recommendation results when new listings matching the user's criteria become available.
FR-26: The system shall allow users to reset or update their preferences without losing previously saved listings.
FR-27: The system shall log user interaction data (clicks, saves, refinements) to improve future recommendation accuracy.
FR-28: The system shall generate a confidence score for each recommendation to indicate how closely it matches user constraints.
FR-29: The system shall provide filtering and sorting options (price, commute distance, safety rating, newest listings).
FR-30: The system shall allow users to directly navigate to the official Zillow listing page to complete applications or view detailed property information.

Abhiram Annavajhula
FR-31: The system shall allow users to create and manage a personal account using email registration and secure login authentication
FR-32: The system shall allow users to edit and update their user profile, including relocation timeline, preferred city, lifestyle priorities, and commuting preferences
FR-33: The system shall provide an interactive map view that displays recommended apartments and neighborhoods geographically
FR-34: The system shall allow users to compare multiple neighborhoods side-by-side based on metrics such as safety score, average rent, commute distance, and available amenities
FR-35: The system shall allow users to schedule apartment tours or request additional information from property managers through an integrated request form
FR-36: The system shall allow users to submit maintenance requests through an AI-powered ticketing system that categorizes the issue type and assigns apriority level
FR-37: The system shall provide status tracking for submitted maintenance requests, allowing users to view updates such as submitted, in progress, or resolved
FR-38: The system shall allow users to receive notifications or alerts when new listing appear that match their saved preferences
FR-39: The system shall allow users to view neighborhood trend insights, including average rent changes and population demographics over time
FR-40: The system shall allow users to export or share selected apartment listings or neighborhood comparisons via a shareable link or downloadable report

2.3 Non-Functional Requirements (NFR)

Rithanya Dhivakar

NFR-01 The system shall load and display recommendation results within 3 seconds after the results page is requested under normal operating conditions.
NFR-02 The system shall regenerate and display updated recommendation results within 2 seconds after a user modifies preferences or submits a preference refinement.
NFR-03 The system shall limit default recommendation explanations to a maximum of three primary influencing factors, with additional explanation details available through optional user interaction.
NFR-04 The system shall apply behavior-based personalization only after detecting a minimum of three consistent user interactions and shall enforce defined fairness and bias-mitigation rules across all generated recommendations.

Jose Rogel
NFR-05 The system must load apartment search results within 2.5 seconds of a user submitting a query under normal network conditions, regardless of the number of active filters applied.
NFR-06 The apartment search, listing, amenity, About, and Contact Us pages must meet WCAG 2.1 AA accessibility standards, ensuring usability for users with disabilities across all devices.
NFR-07 The apartment search, listing, and amenity pages must render consistently and remain fully functional across all major browsers including Chrome, Safari, Firefox, and Edge on both mobile and desktop.
NFR-08 Amenity and neighborhood data displayed on listing pages must reflect updates from third-party sources within 24 hours via a scheduled sync, and all listings must display a "Last Updated" timestamp so users are aware of data freshness.

Abhiram Annavajhula
NFR-09: The system shall support at least10,000 concurrent users without significant degradation in system performance
NFR-10: The system shall allow new users to complete the initial preference setup within 5 minutes without external assistance
NFR-11: The system interface shall allow new users to complete the initial preference setup within 5 minutes without external assistance
NFR-12: The system shall automatically recover from temporary server failures and restore service within 5 minutes without requiring manual intervention

Shreya Sil
NFR-13: The system shall generate updated recommendations within 3 seconds of user input submission.
NFR-14: The system shall encrypt all user data both in transit and at rest.
NFR-15: The system shall maintain at least 99% uptime to ensure consistent system access.
NFR-16: The interface shall allow users to generate recommendations in no more than three steps.

2.4 Agile / Scrum User Stories

Rithanya Dhivakar

US-01 As a relocating renter, I want to clearly enter and review my housing preferences, and adjust what I care about when needed, so the recommendations reflect my priorities without requiring me to start over.

US-02 As a relocating renter, I want to receive recommendations that reflect my strongest constraints and include clear explanations and trade-offs, so I can understand why options are suggested and make informed decisions.

US-03 As a relocating renter, I want recommendations to be presented with an appropriate level of guidance based on my rental experience, and to explore additional details only when I choose, so the information feels useful without being overwhelming.

US-04 As a relocating renter, I want my recommendations to improve based on my interactions and to receive helpful alternatives when no strong matches exist, so I can continue my search even when my preferences are strict.

Jose Rogel

US-05 As a prospective renter, I want to search for apartments by location, keyword, or zip code and filter results by cost, bedrooms, and commute distance, so I can quickly narrow down listings that fit my priorities without sorting through irrelevant options.

US-06 As a prospective renter, I want to view a detailed apartment listing with amenities, safety score, nearby points of interest, and a direct way to contact the property, so I can evaluate whether a unit fits my lifestyle before reaching out.

US-07 As a prospective renter, I want to save apartment listings to a favorites list that persists across my devices, so I can compare my top choices later without having to search for them again.

US-08 As a prospective renter, I want to easily access the platform's About, FAQ, Privacy Policy, and Terms of Service pages, so I can understand how Settle works, how my data is used, and who to contact if I need help.

Abhiram Annavajhula

US-09: As a relocating renter, I want to view recommended apartments on an interactive map, so that I can understand their location relative to workplaces, transit, and nearby neighborhoods

US-10: As a renter, I want to receive alerts when new apartments become available that match my preferences, so that I don't miss any good opportunities

US-11: As a tenant, I want to submit maintenance requests through the platform, so that I can report issues with my apartment quickly and easily

US-12: As a renter, I want to share apartment listings with friends, family, or roommates, so that I can get feedback before deciding.

Shreya Sil

US-13: As a relocating renter, I want to enter my budget and prefer commute distance so that I can receive housing recommendations within my constraints.

US-14: As a user, I want to modify my preferences after seeing results so that I can refine my recommendations.
US-15: As a renter moving to a new city, I want to view neighborhood insights (safety, amenities, transit access) so that I feel confident about where I choose to live.
US-16: As a Zillow administrator, I want user interaction data to be collected so that the recommendation system can continuously improve accuracy.

2.5 Stakeholder Table

Role	Company	Questions	Answers	Owner
Product Marketing Director	Zillow	How should the search results adapt to different customer segments, such as first-time buyers versus real-estate investors?	Search results should emphasize guidance and confidence for first-time buyers, with clearer explanations, fewer choices, and educational cues. For seasoned investors, results should surface comparative metrics, risk indicators, and performance trends, enabling faster scanning and data-driven decisions without extra handholding.	Rithanya Dhivakar
Product Marketing Director	Zillow	When a user lands on the results page, what's the most important takeaway we want them to understand within a few seconds?	The user should immediately understand why these results fit their needs. The page should clearly signal relevance, showing that their preferences were heard and translated into	Rithanya Dhivakar

			meaningful recommendations, so they feel confident continuing rather than second-guessing the system.	
Product Marketing Director	Zillow	What level of explanation helps users trust the recommendation without feeling like the system is deciding for them?	High-level, plain-language reasons work best. Users should see a few clear factors that influenced the recommendation, with the option to explore deeper. This keeps them informed and in control, while avoiding the feeling that the system is making decisions on their behalf.	Rithanya Dhivakar
Data Scientist	Zillow	Which user preferences tend to be the most reliable predictors of satisfaction or successful outcomes?	Budget range and commute tolerance are the most reliable because users consistently act on them. Safety and neighborhood vibe matter, but they're more subjective. Preferences tied to real constraints produce more stable models and stronger signals of satisfaction over time.	Rithanya Dhivakar
Data Scientist	Zillow	How can free-text preference input be supported without	Free-text input should be interpreted as a	Rithanya Dhivakar

		weakening the reliability of structured user preference data?	refinement layer that adjusts existing structured preferences rather than replacing them. This preserves data stability while allowing users to express changing priorities in a flexible but controlled way.	
Data Scientist	Zillow	How do you balance personalization with fairness constraints in housing-related recommendations?	We separate user intent from protected attributes and apply fairness constraints at the model and post-processing levels. Personalization focuses on preferences like price and commute, while guardrails ensure recommendations don't systematically exclude neighborhoods or reinforce biased outcomes.	Rithanya Dhivakar
Data Scientist	Zillow	At what point do you start adjusting recommendations based on user behavior instead of stated preferences?	We start blending behavior once we see consistent signals, like repeated clicks or saves across sessions. Stated preferences anchor early recommendations, but behavioral data gradually takes	Rithanya Dhivakar

			more weight as it proves more reliable than initial self-reported inputs.	
Business Operations Lead	Zillow	When should the system begin incorporating user interaction behavior into recommendation adjustments?	Behavioral signals should influence recommendations only after multiple consistent interactions, such as repeated views or saves, to avoid overreacting to isolated actions while maintaining alignment with the user's stated preferences.	Rithanya Dhivakar
Business Operations Lead	Zillow	How should the system respond when a user's preferences are too restrictive to generate strong recommendations?	The system should clearly communicate why strong matches are unavailable and suggest practical alternatives, such as relaxing constraints or exploring nearby areas, to help users continue their search without frustration.	Rithanya Dhivakar
Business Operations Lead	Zillow	How should the system surface cases where no strong recommendations exist?	The system should clearly explain why strong matches aren't available and suggest actionable adjustments, like relaxing constraints or exploring nearby areas. From an ops view, these cases should be flagged	Rithanya Dhivakar

			on the dashboard so teams can identify inventory gaps or overly restrictive filters.	
Product Manager	Zillow	What business goals should the recommendation system prioritize (user engagement, lease conversions, revenue growth, or user satisfaction)?	The recommendation system should prioritize increasing user engagement and improving lease conversions while also supporting revenue growth. At the same time, the system must maintain high user satisfaction by recommending housing options that closely match user preferences such as location, budget, and amenities.	Shreya Sil
Product Manager	Zillow	How should the system balance personalized recommendations with fairness and legal compliance requirements?	The system should balance personalization with fairness by ensuring that recommendations are not biased or discriminatory. It should follow legal housing regulations and avoid filtering results based on protected characteristics. The algorithm should focus only on relevant user preferences like price range,	Shreya Sil

			location, and housing features.	
Data Scientist	Zillow	What data sources are most reliable for generating accurate and unbiased housing recommendations?	The most reliable data sources include verified property listings, user preference data, historical rental trends, and market pricing data. Using multiple trusted data sources helps ensure the recommendations are accurate, current, and unbiased.	Shreya Sil
Data Scientist	Zillow	How should we measure the effectiveness and accuracy of the recommendation algorithm over time?	We can measure effectiveness using metrics such as click-through rates on recommended listings, successful lease conversions, and user satisfaction feedback. Over time, analyzing these metrics helps improve the recommendation algorithm and ensure it continues to provide useful suggestions.	Shreya Sil
Business Operations Lead	Zillow	How should the system respond if a user's preferences are too restrictive and result in no matches?	If a user's preferences are too restrictive and produce no matches, the system should suggest relaxing certain filters such	Shreya Sil

			as price range or distance. It can also recommend similar listings that are slightly outside the user's criteria to help them discover more options.	
Business Operations Lead	Zillow	What processes should be in place to update recommendation rules as rental market conditions change?	There should be a process for regularly updating recommendation algorithms based on changing rental market conditions. This includes updating pricing data, adding new listings, removing unavailable properties, and adjusting recommendation models to reflect current housing trends.	Shreya Sil
Legal/Compliance Officer	Zillow	What safeguards must be implemented to ensure compliance with Fair Housing regulations?	Safeguards should include ensuring the system does not discriminate against users based on protected characteristics such as race, religion, gender, or family status. The system must comply with Fair Housing regulations by focusing recommendations only on valid housing factors like	Shreya Sil

			location, price, and property features.	
Legal/Compliance Officer	Zillow	What user data must be protected, anonymized, or restricted to prevent misuse?	User data such as personal information, search history, and preferences should be protected through encryption and secure storage. Sensitive information should be anonymized when used for analytics to prevent misuse and protect user privacy.	Shreya Sil
End User (Renter)	Zillow	What level of explanation would make you trust the system's recommendations?	Users would trust the system more if it clearly explains why certain properties are recommended, such as matching price range, preferred location, or selected amenities. Providing transparency helps users understand how the recommendations were generated.	Shreya Sil
End User (Renter)	Zillow	How much control would you like over modifying or overriding AI-generated suggestions?	Users should have the ability to modify filters, adjust preferences, and override AI-generated recommendations. This control allows users to refine their search and ensures the system remains	Shreya Sil

			helpful rather than restrictive.	
Relocation Specialist	Zillow	What information do people relocating to a new city typically wish they had known before choosing a neighborhood?	This question identifies information gaps for relocating renters, such as safety, commute time, and nearby amenities. These insights generated requirements for displaying neighborhood insights and local data.	Abhiram Annavajhula
Urban Planner	Zillow	Which neighborhood-level indicators, such as walkability, transportation, access, or zoning, would provide meaningful context for someone moving to a new city?	This question determines which neighborhood indicators (walkability, transit access, zoning) help users evaluate locations. It generated requirements for displaying neighborhood metrics.	Abhiram Annavajhula
Property Manager	Zillow	What types of maintenance issues are most commonly reported by renters, and how should the system categorize them?	This question identifies common renter maintenance issues like plumbing or electrical problems. It generated requirements for a categorized maintenance request system.	Abhiram Annavajhula
Customer Support Manager	Zillow	What are the most frequent questions or problems renters	This question helps identify This question identifies	Abhiram Annavajhula

		encounter when using housing platforms?	common renter pain points. It generated requirements for clear listings, accurate data, and easy communication features.	
UX Designer	Zillow	What information should be visible to a user when they first compare two neighborhoods so the comparison is quick and meaningful	This question identifies key comparison factors like rent, safety, and commute time. It generated requirements for side-by-side neighborhood comparison.	Abhiram Annavajhula
Security Engineer	Zillow	What authentication or account protection features should be implemented protect user profiles and personal housing preferences?	This question focuses on protecting user data and accounts. It generated requirements for secure login and account protection features.	Abhiram Annavajhula
Real Estate Market Analyst	Zillow	What market trends or rental indicators should the platform monitor to help users understand whether a neighborhood's rent prices are rising or stable	This question identifies useful housing market indicators like rent trends. It generated requirements for tracking and displaying rental market trends.	Abhiram Annavajhula
Community Engagement Coordinator	Zillow	How could the platform incorporate community feedback or resident reviews to help users better understand	This question explores how community feedback can describe neighborhood culture. It generated	Abhiram Annavajhula

		neighborhood culture?	requirements for resident reviews or ratings.	
Platform Analytics Manager	Zillow	What user behavior metrics should we track to understand how effectively users are finding neighborhoods that match their preferences	This question identifies important analytics such as searches, views, and saved listings. It generated requirements for tracking user behavior metrics.	Abhiram Annavajhula
Mobile/Web Accessibility Specialist	Zillow	What design considerations would ensure that users with different abilities can easily navigate neighborhood data and recommendation results	This question focuses on accessibility and inclusive design. It generated requirements for accessible and easy-to-navigate interfaces.	Abhiram Annavajhula
Product Lead	Zillow	What are the most common ways to search for any apartment?	Users almost always start with location such as city, zip, or neighborhood followed immediately by a strict maximum budget. For young professionals, commute time to a specific workplace and pet-friendly policies are also high-priority entry points.	Jose Rogel
Lead UX Researcher	Zillow	What options to filter are most important to renters when reducing down listings?	Renters prioritize hard limits like total monthly cost including utilities and lifestyle	Jose Rogel

			logistics like safety heatmaps or proximity to specific points of interest. The ability to toggle these priorities via sliders is essential for personalization.	
Product Lead	Zillow	What information must be visible for a user to feel confident enough to reach out?	Radical transparency is required. This includes a full total cost breakdown covering rent plus mandatory fees, unit-specific media rather than model photos, and a direct messaging feature to avoid phone anxiety.	Jose Rogel
CMO	Zillow	How much weight do renters place on walkability and specific amenities?	Walkability is a top-three priority. The target demographic views the apartment as a social home base, placing extreme value on proximity to third spaces like climbing gyms, trails, and local coffee shops.	Jose Rogel
Lead UX Researcher	Zillow	Should the platform support saving or tracking listings across sessions?	Yes. The average renter in this bracket takes two to four weeks to decide and visits top choices at least five times. Persistent accounts and a comparison view are core requirements to	Jose Rogel

			support this multi-session behavior.	
Legal & Compliance Lead	Zillow	Are there accessibility standards or government compliance requirements to meet?	The platform must adhere to the Fair Housing Act, WCAG 2.1 Level AA for ADA accessibility, and the Texas Data Privacy and Security Act for user data encryption.	Jose Rogel
Head of Engineering	Zillow	What is the acceptable threshold for search response time?	The abandonment cliff is 3 seconds. Users expect load times of 2 seconds or less, and interactive elements like the Compatibility Tracker must respond within 200 milliseconds to feel functional.	Jose Rogel
Product Lead	ZIllow	How often does amenity or neighborhood data change in practice?	Price and availability are highly volatile and require a sync at least every 24 hours, ideally every few hours. Lifestyle amenities change weekly, while safety metrics are typically updated on a quarterly cycle.	Jose Rogel
Systems Administrator	Zillow	Is it better to sync every 24 hours for better reliability?	While a 24-hour batch sync is more stable for the database, it risks losing user trust. A delta sync is recommended, syncing only price and status changes	Jose Rogel

			hourly while performing full refreshes of static data every 24 hours.	
Legal & Compliance Lead	Zillow	What legal agreements must a user acknowledge before the platform can create their account?	Users must explicitly acknowledge both a Terms of Service agreement and a Privacy Policy before account creation can be completed. These pages must be accessible at any point from the platform and must clearly outline user rights, data usage, and platform liability in plain language.	Jose Rogel
Head of Engineering	Zillow	What browser and device compatibility requirements must the platform meet to serve the full range of target users?	The platform must render consistently and remain fully functional across Chrome, Safari, Firefox, and Edge on both mobile and desktop. Given that the target demographic of young adults relocating to Dallas is likely to access the platform across multiple device types, cross-browser stability is a core technical requirement rather than an optional enhancement.	Jose Rogel

2.6 Site Map

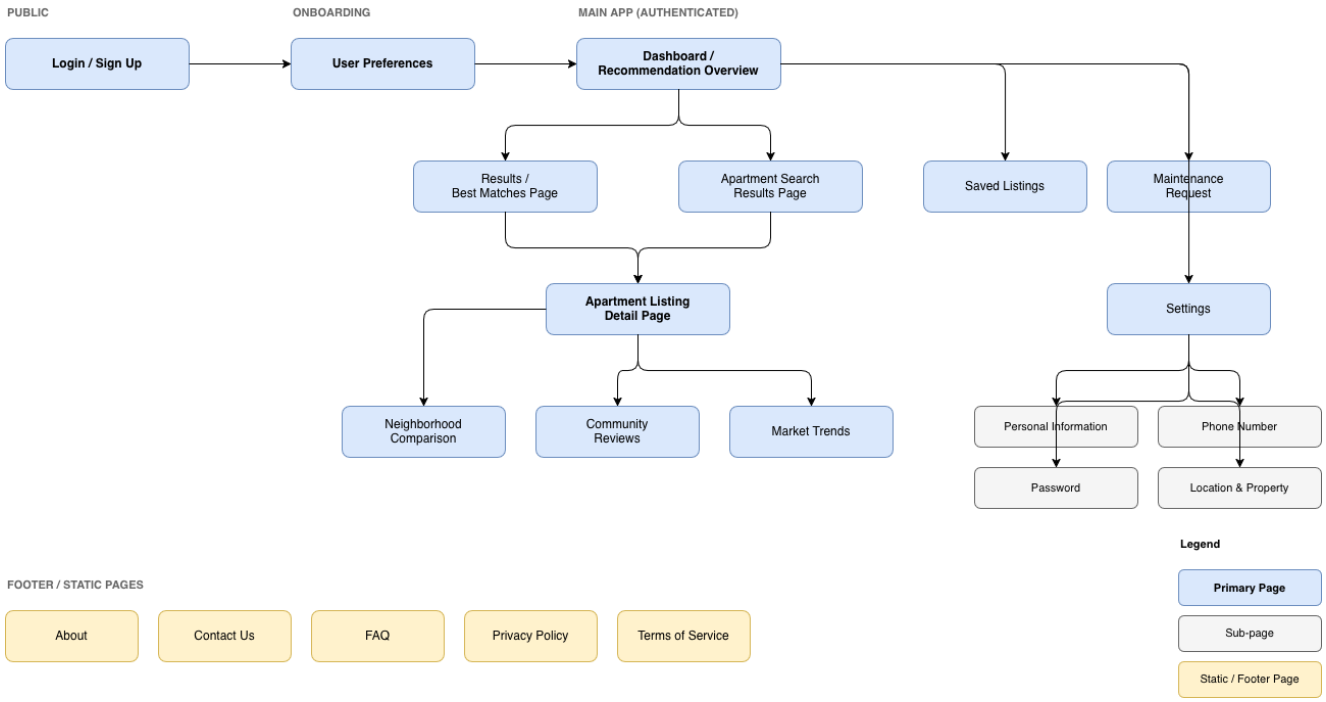

Figure 1 - Site Map by Jose Rogel

2.7 Use Case Diagrams

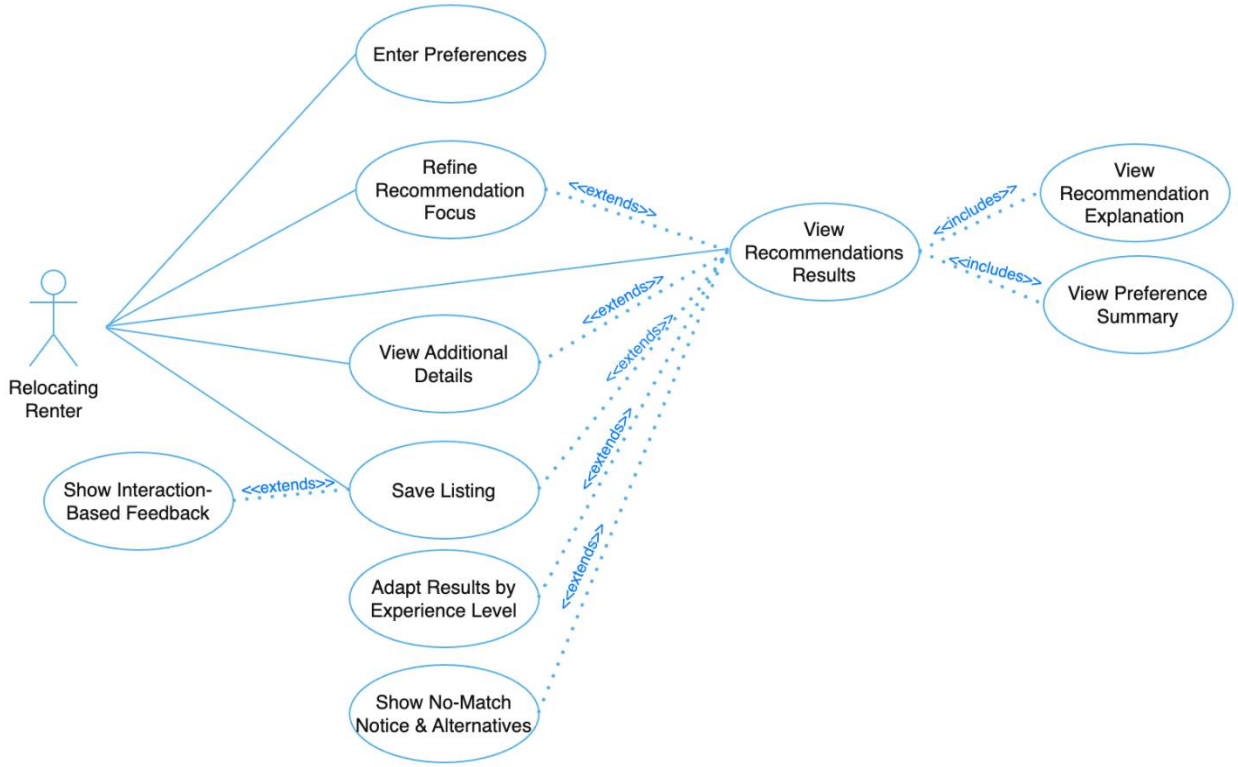

Figure 2 - Rental Recommendation Interaction System Diagram by Rithanya Dhivakar

Figure 2 models the interactions among a relocating renter and the Settle rental recommendations system, highlighting how the user can enter preferences, view recommendations and explanations, and how results are refined or adapted based on user interactions and experience level.

Figure 3 – Prospective Rental System Diagram by Jose Rogel

Figure 3 models the interactions between a prospective renter and the Settle apartment listing system, illustrating how a user can search for and view listings, explore nearby amenities, assess neighborhood safety, contact a property, and save listings to their favorites.

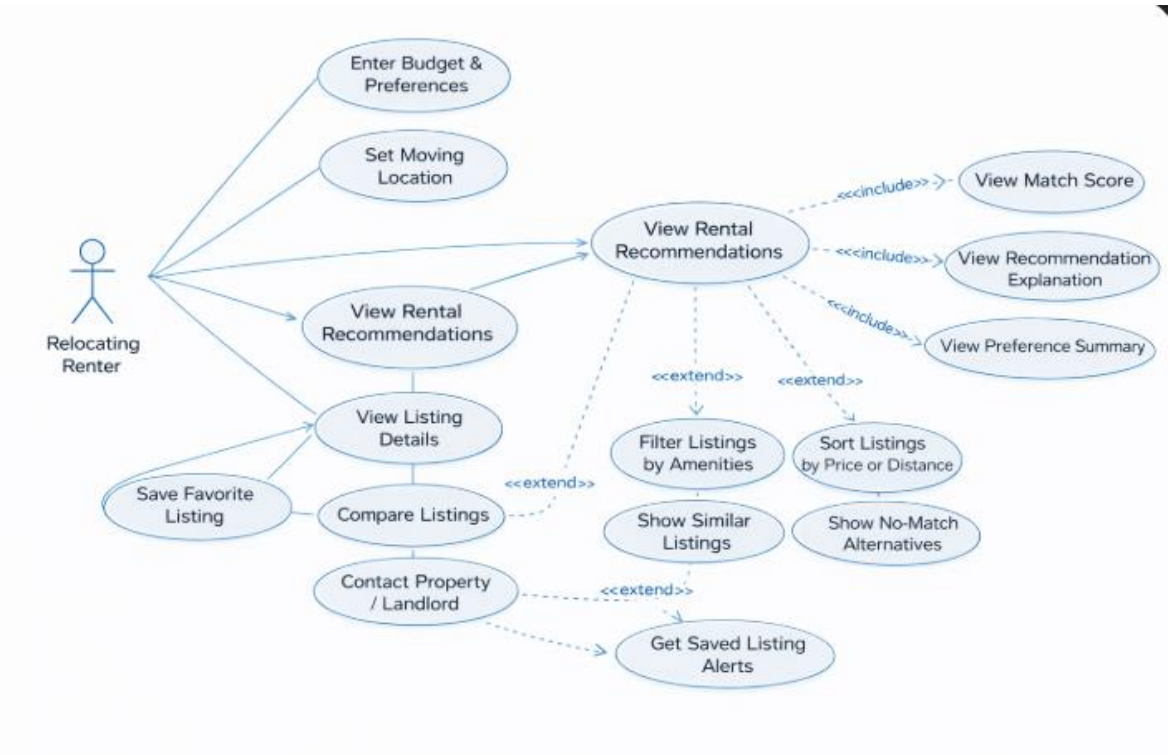

Figure 4 – Relocating Rental System Diagram by Shreya Sil

Figure 4 models the interactions between a relocating renter and the Settle rental recommendation system, illustrating how a user can enter housing preferences, view personalized rental recommendations, explore listing details, compare properties, and save listings. The diagram also highlights additional system features such as filtering results, viewing recommendation explanations, and receiving alerts for saved listings.

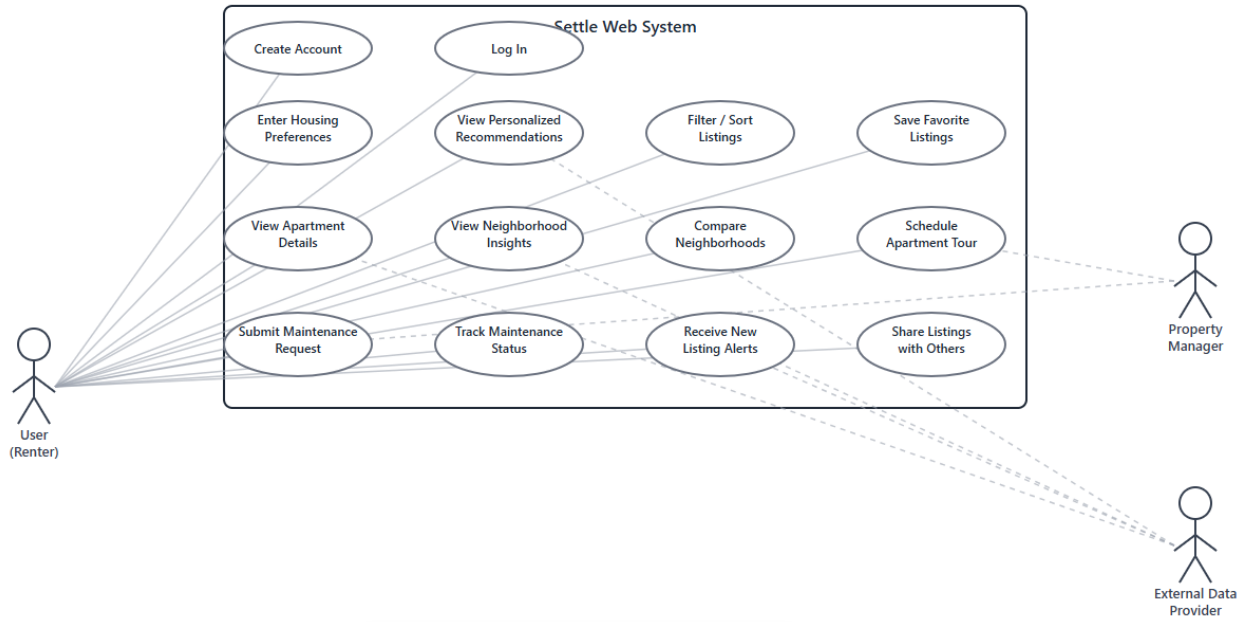

Figure 5 – User-Web Application Interaction System Diagram by Abhiram Annavajhula

Figure 5 models the use case diagram, which represents the main interactions between users and the web application, showing how different actors interact with the system to accomplish tasks related apartment searching and rental decision-making.

2.8 Sequence Diagrams

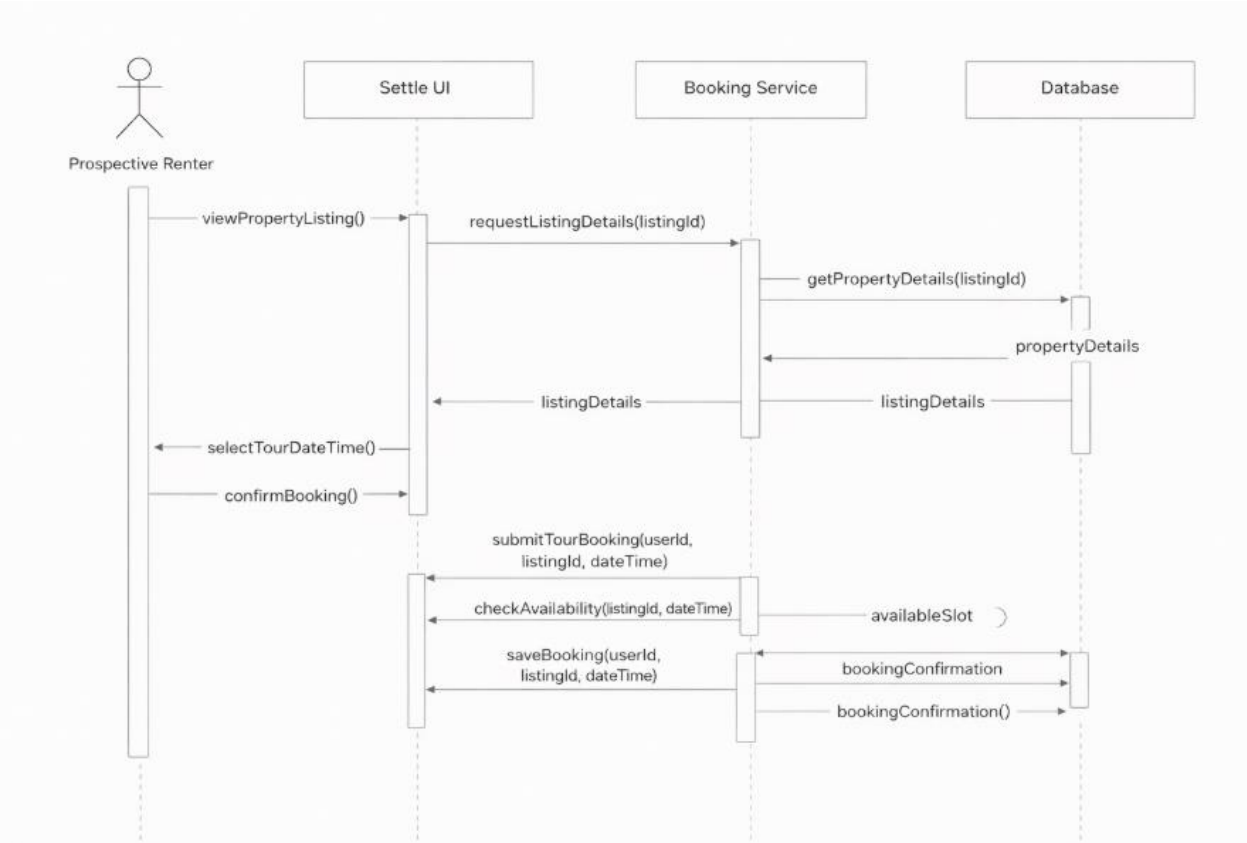

Figure 6 – Booking Interaction Flow Diagram by Shreya Sil

Figure 6 illustrates the booking interaction flow within the Settle apartment listing system. The diagram shows how a renter selects a property, views available booking options, confirms the booking details, and receives confirmation from the system.

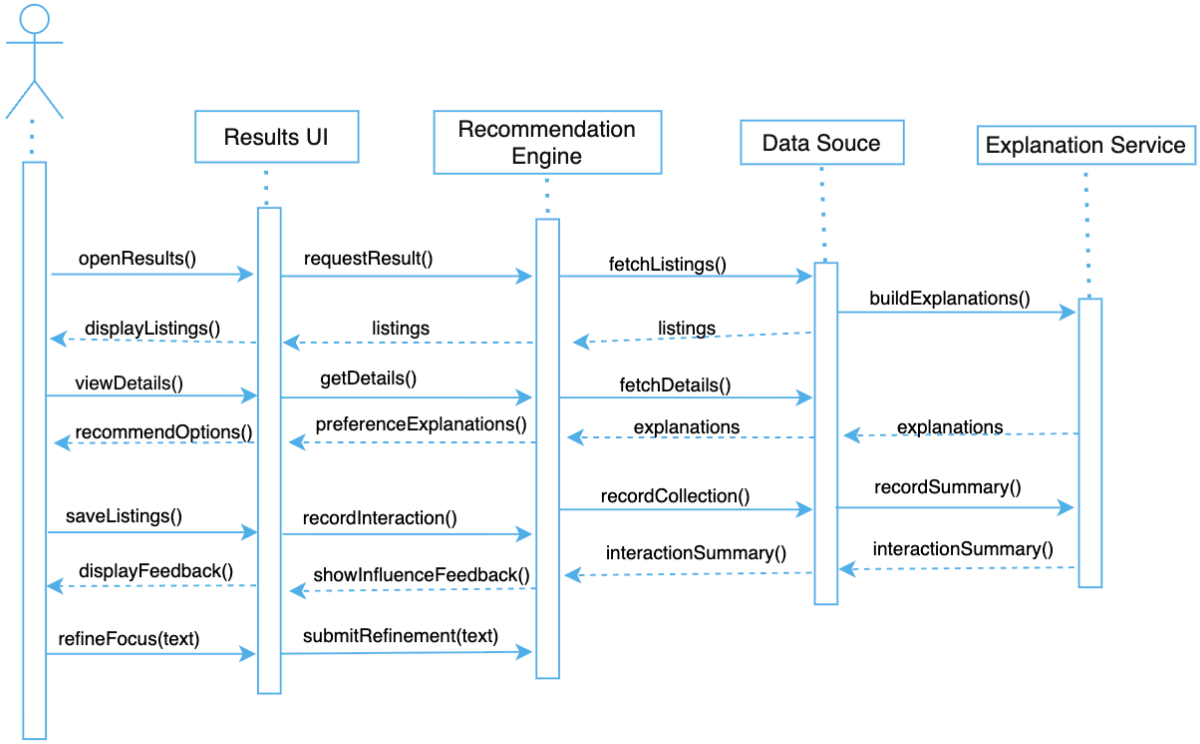

Figure 7 - Recommendation Interaction Flow Diagram by Rithanya Dhivakar

Figure 7 depicts the interaction flow between the user interface and backend services as the system retrieves listings, generates explanations, and updates recommendation results based on user actions.

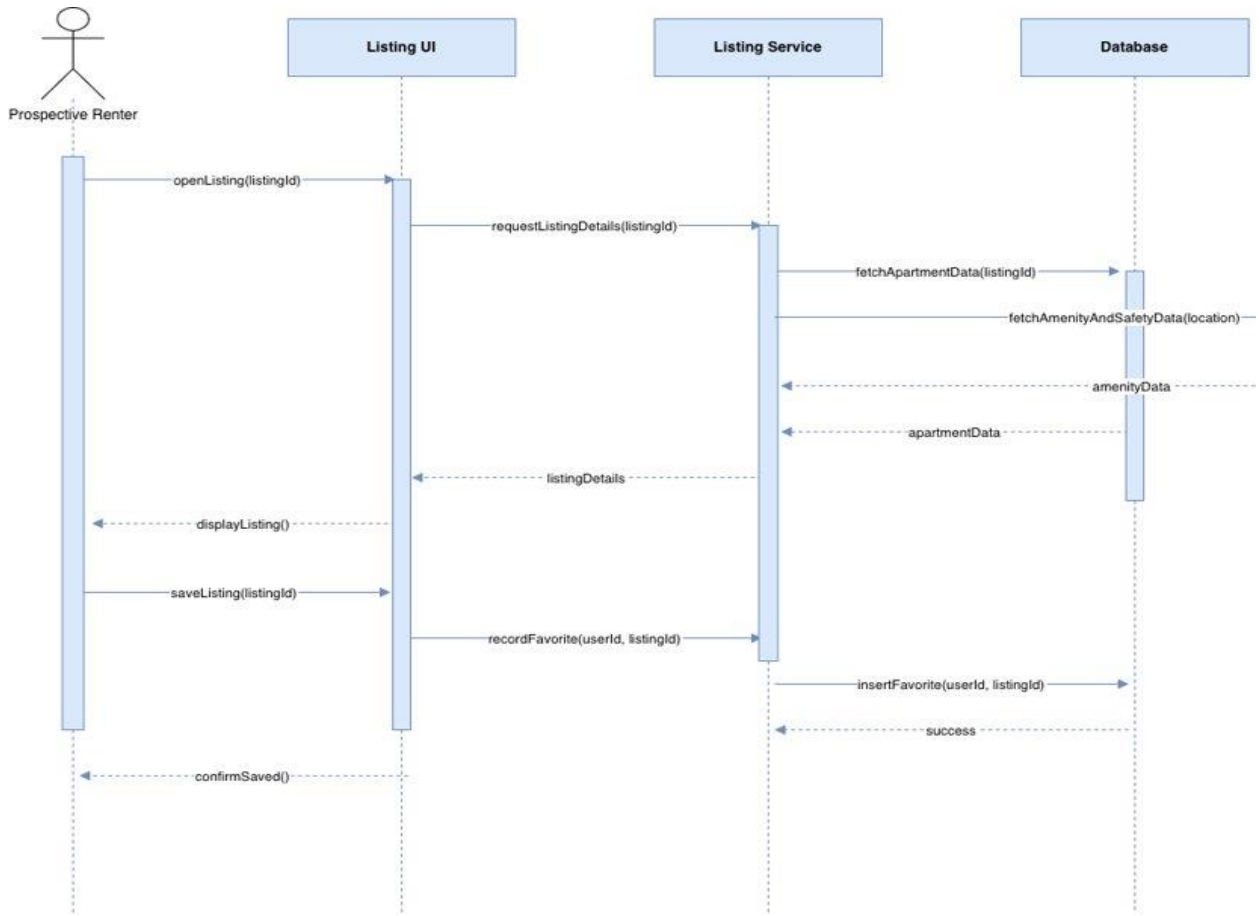

Figure 8 - Apartment Listing Interaction Flow Diagram by Jose Rogel

Figure 8 depicts the interaction flow between the user interface, listing service, database, and data sync service as the system retrieves apartment details, fetches amenity and safety data, and records a user's saved listing.

3. Wire Frames

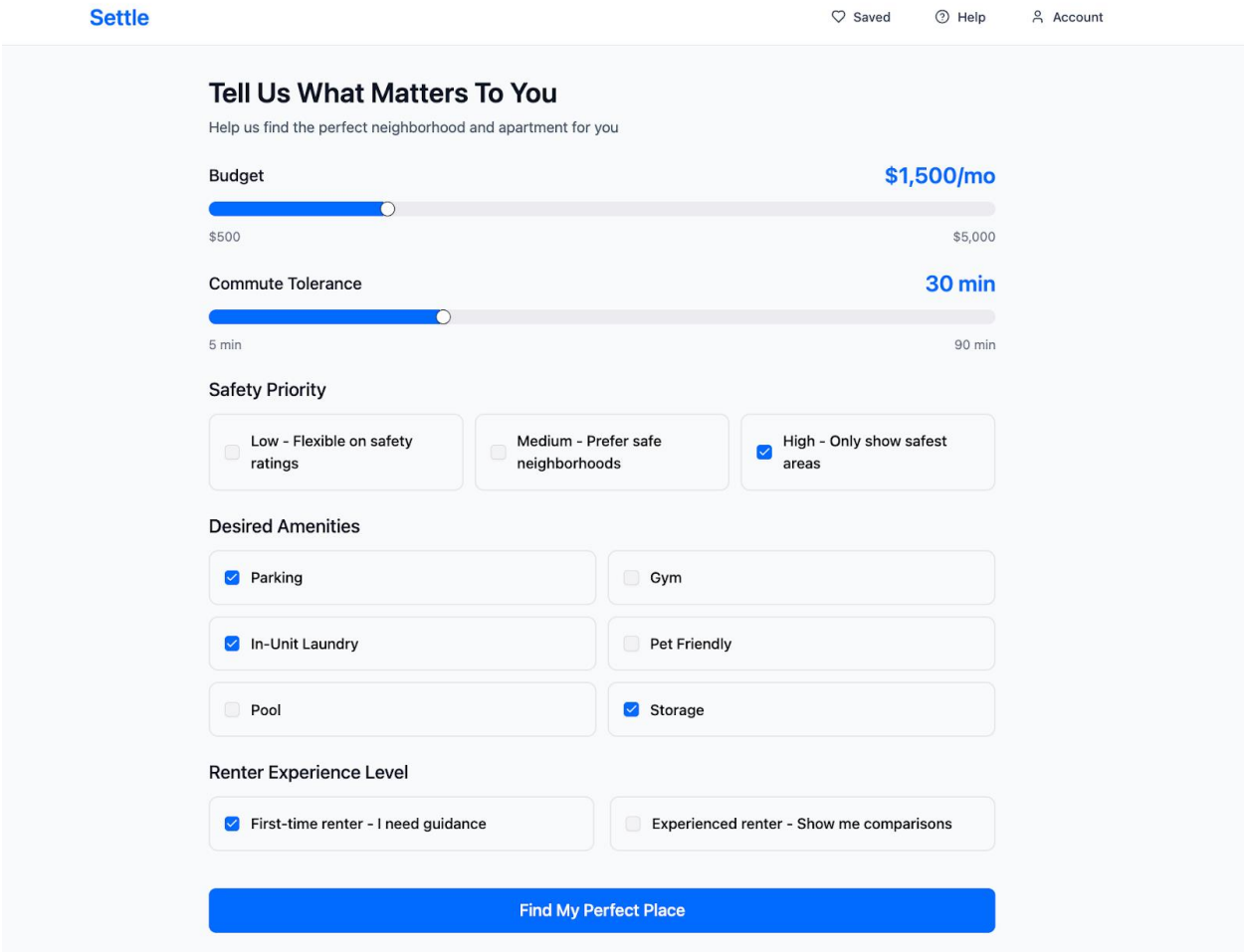

Figure 9 - User Preferences Page by Rithanya Dhivakar

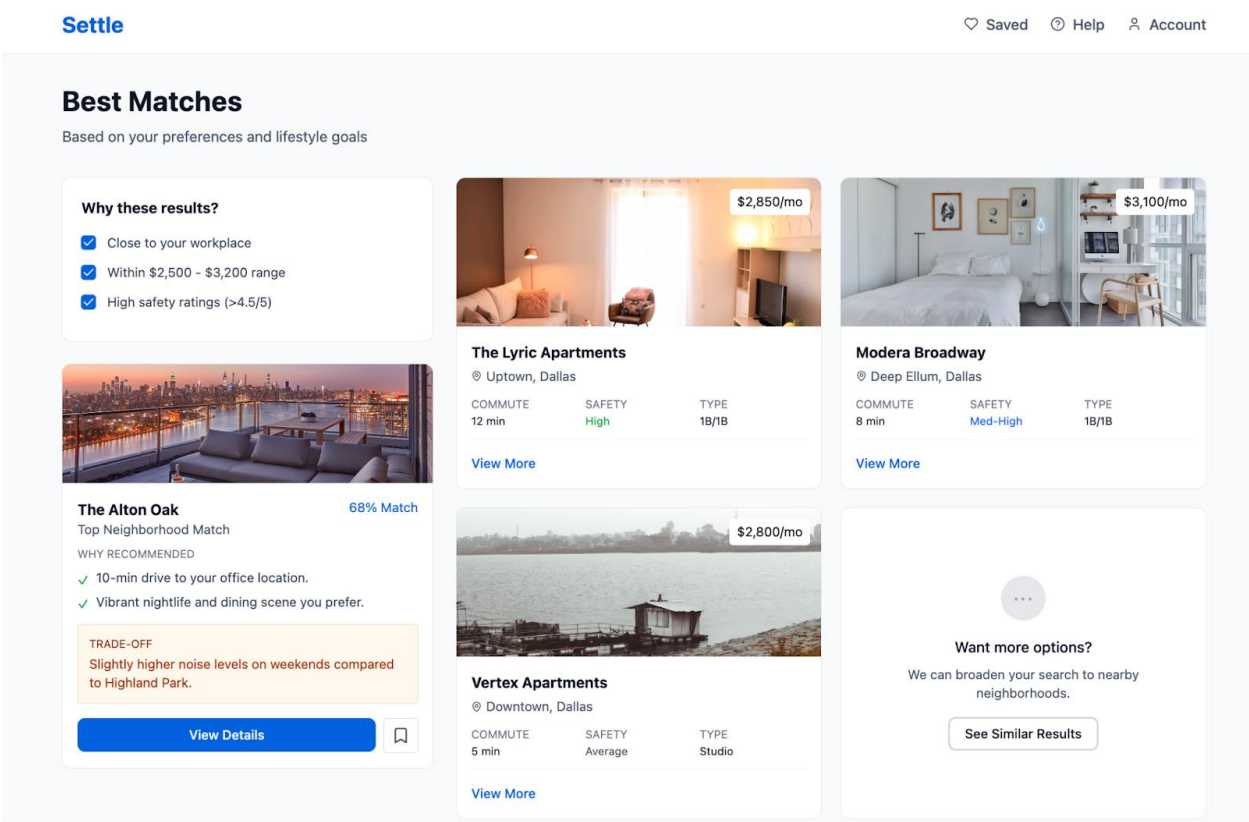

Figure 10 - Preference Results Page by Rithanya Dhivakar

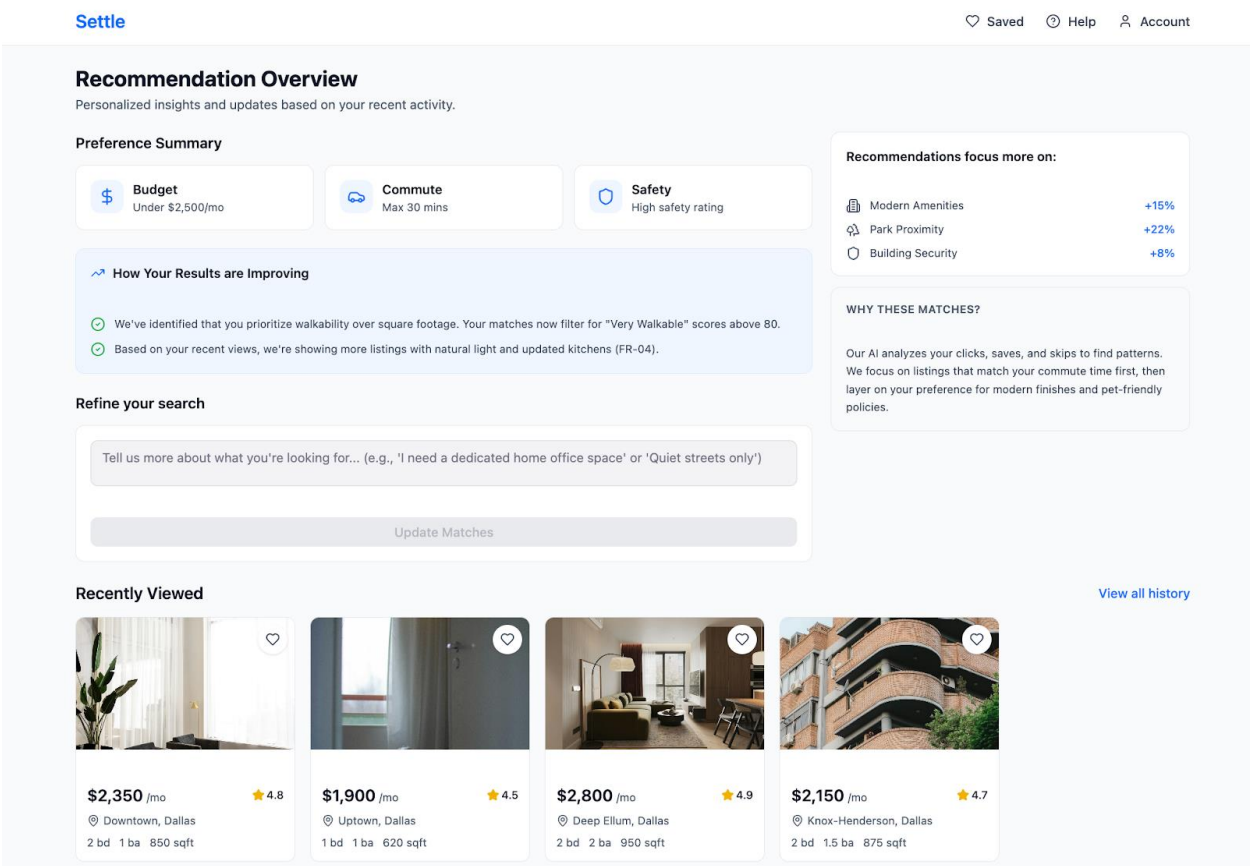

Figure 11 – Dashboard / Recommendation Page by Rithanya Dhivakar

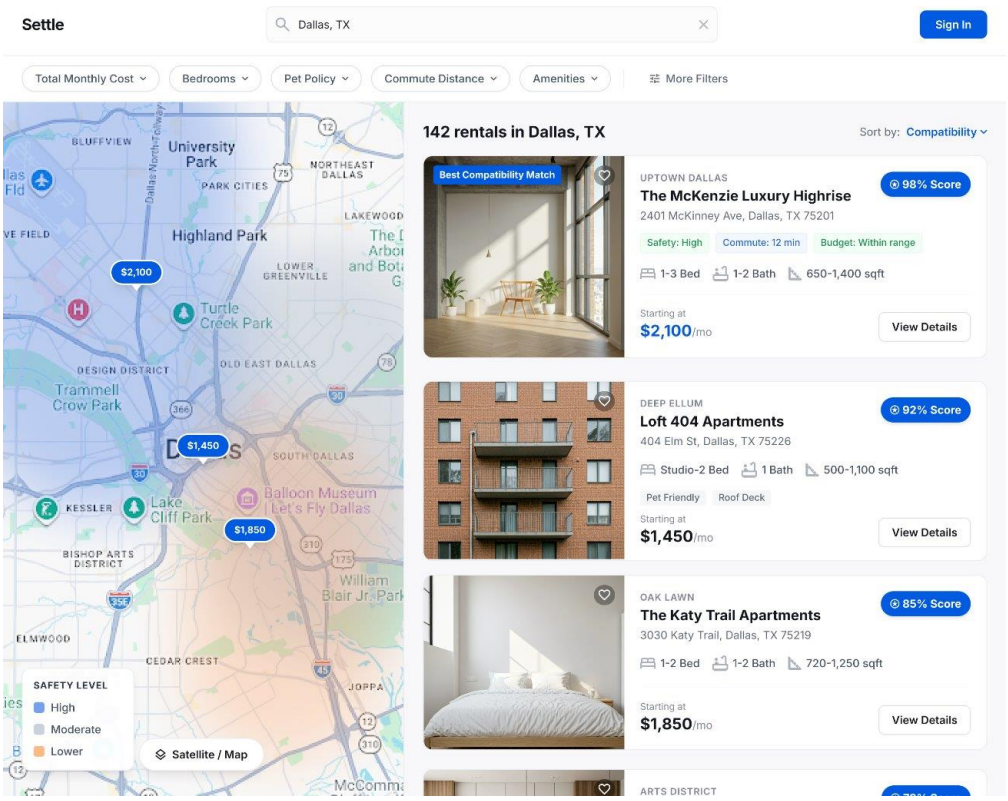

Figure 12 - Apartment Search Results Page by Jose Rogel

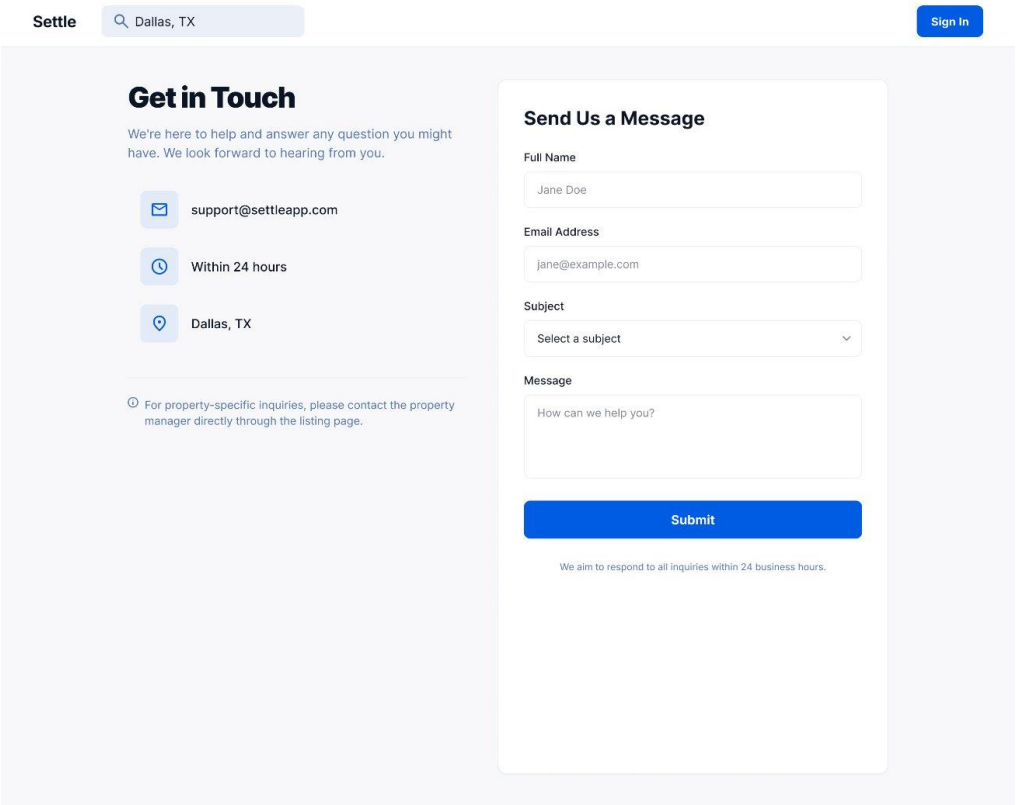

Figure 13 - Contact Us Page by Jose Rogel

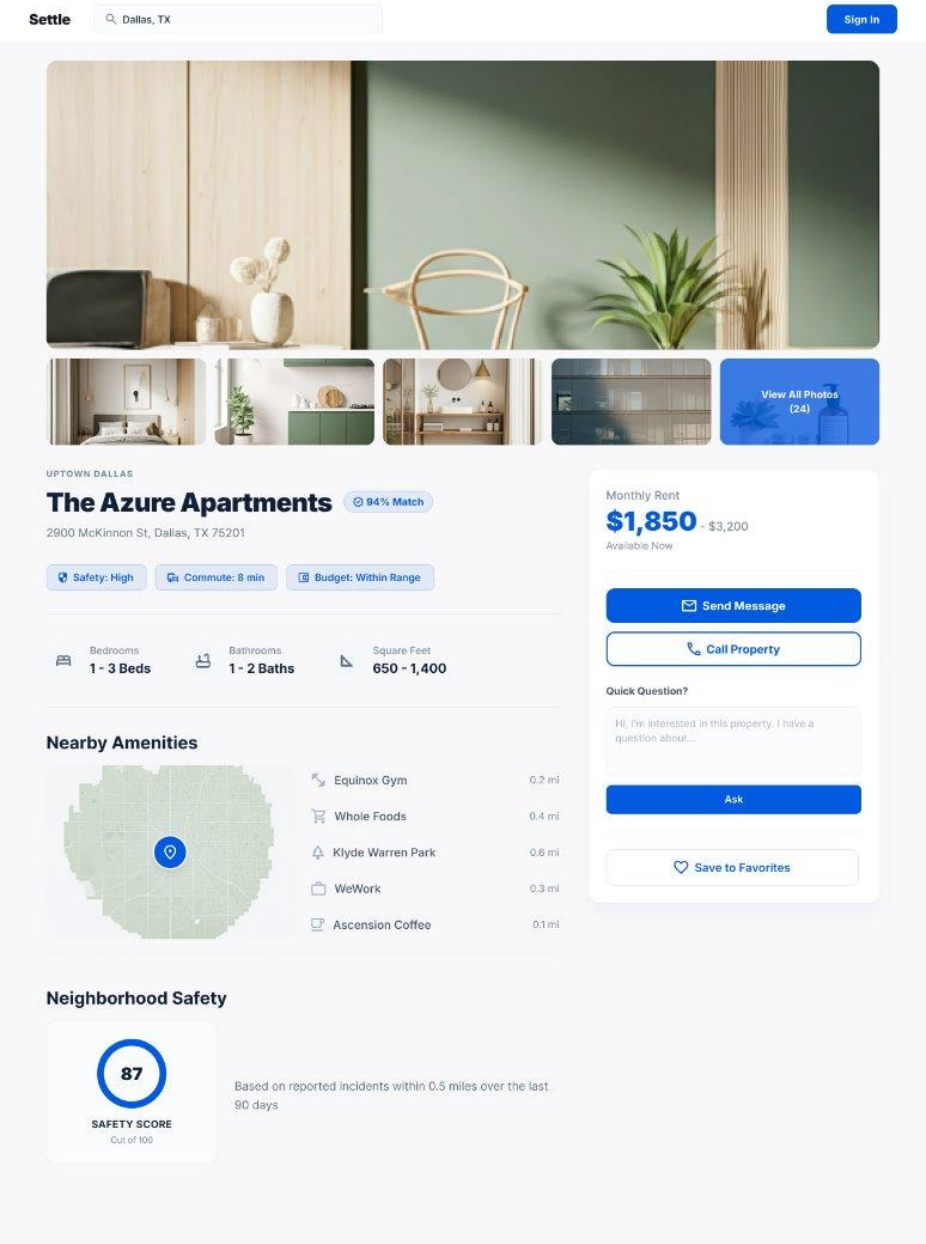

Figure 14 - Apartment Listing Detail Page by Jose Rogel

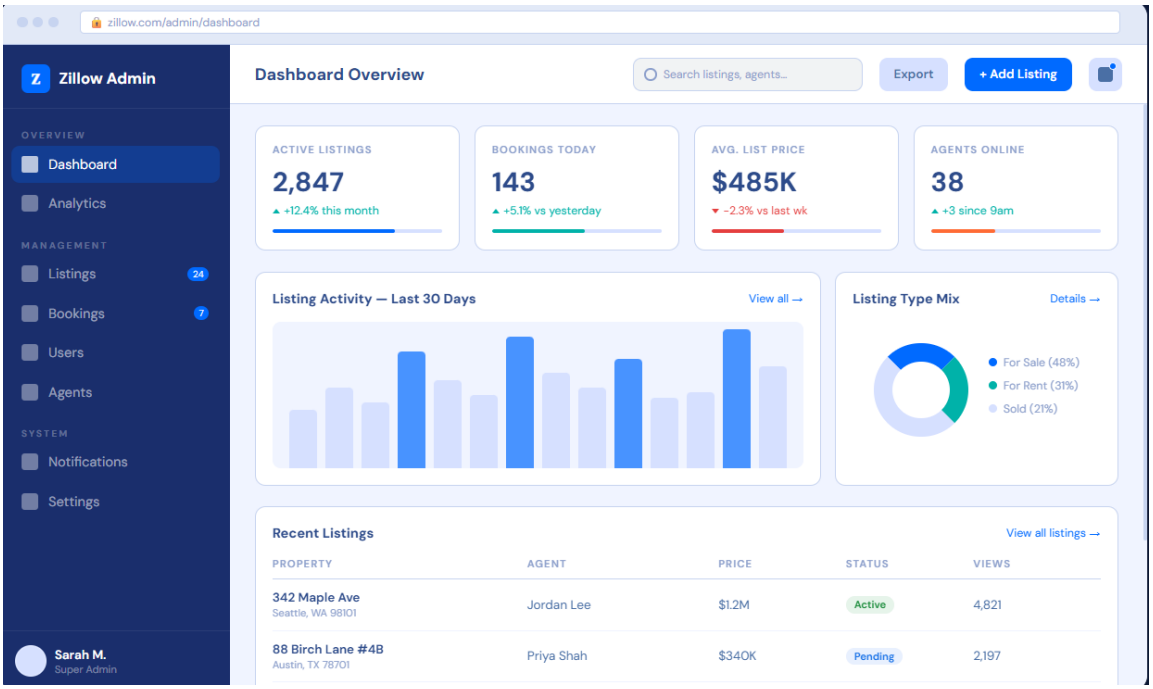

Figure 15 - Admin Dashboard Page by Shreya Sil

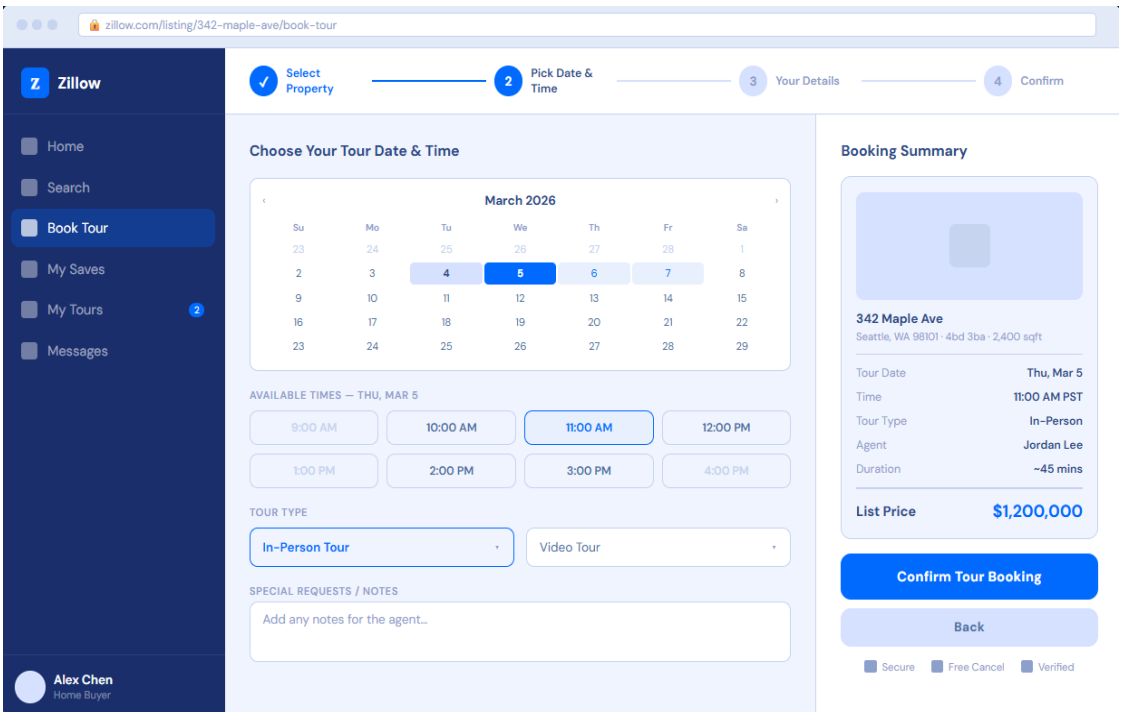

Figure 16 - Booking/Confirmation Page by Shreya Sil

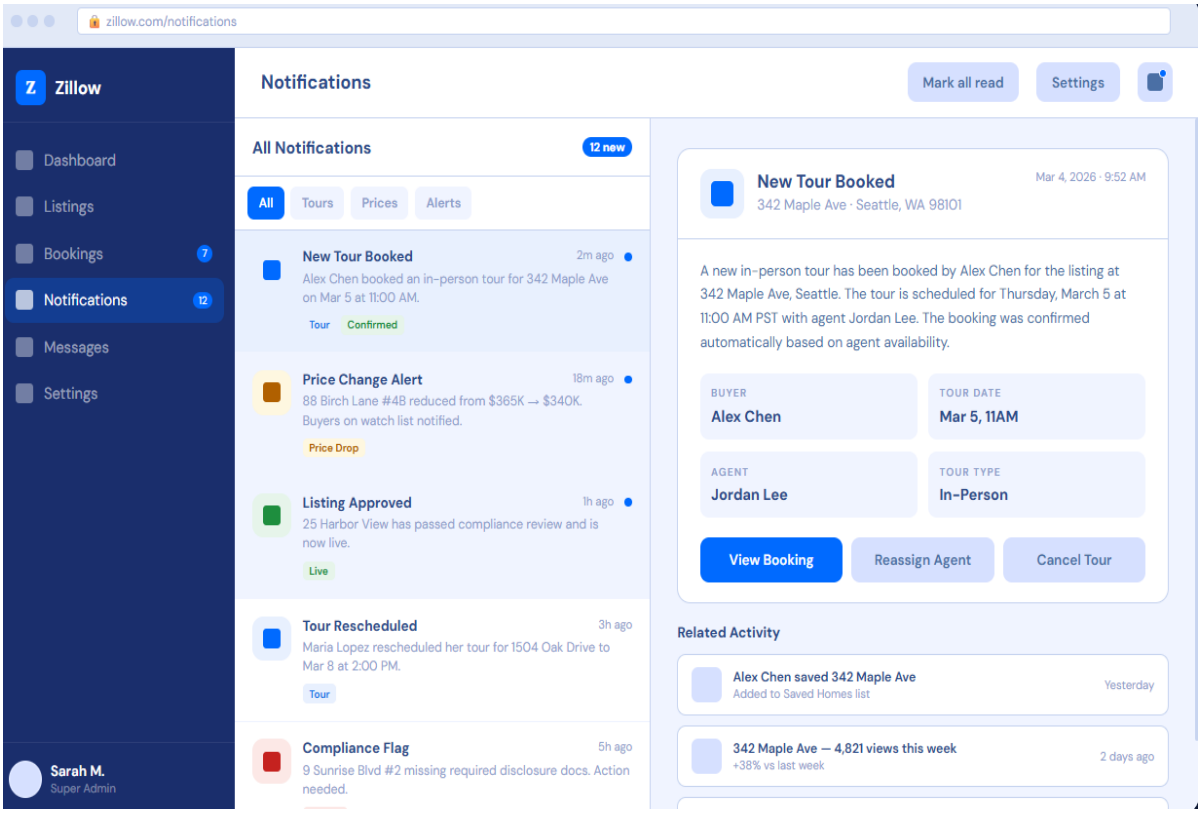

Figure 17 - Notifications Page by Shreya Sil

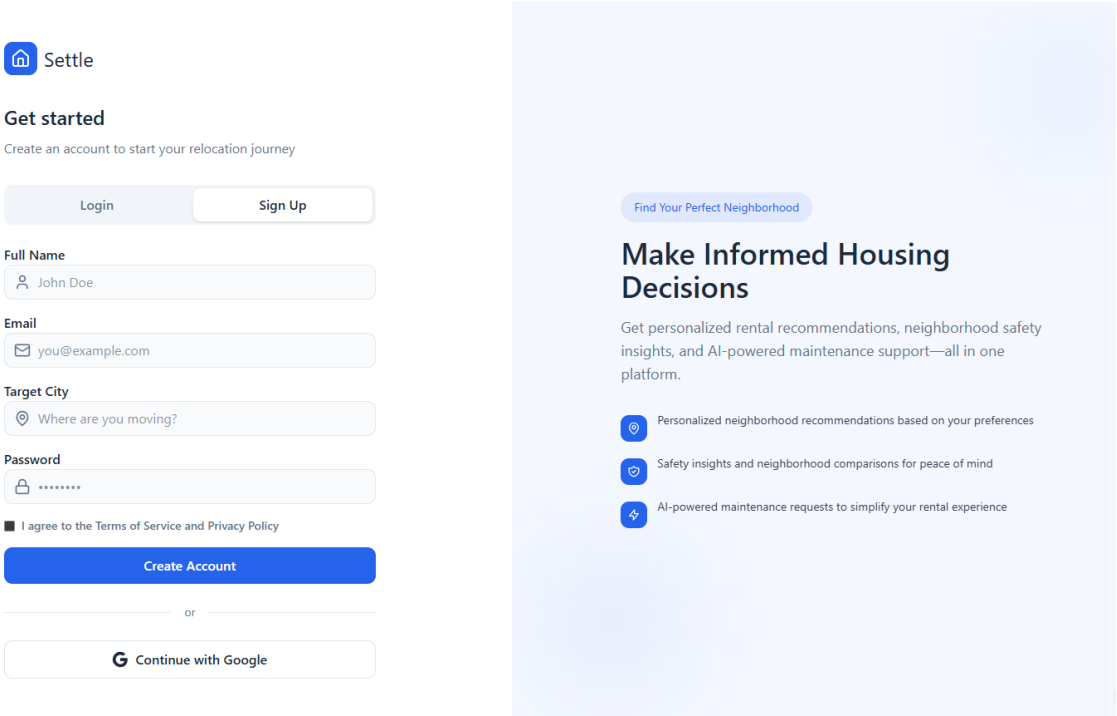

Figure 18 - Login Page by Abhiram Annavajhula

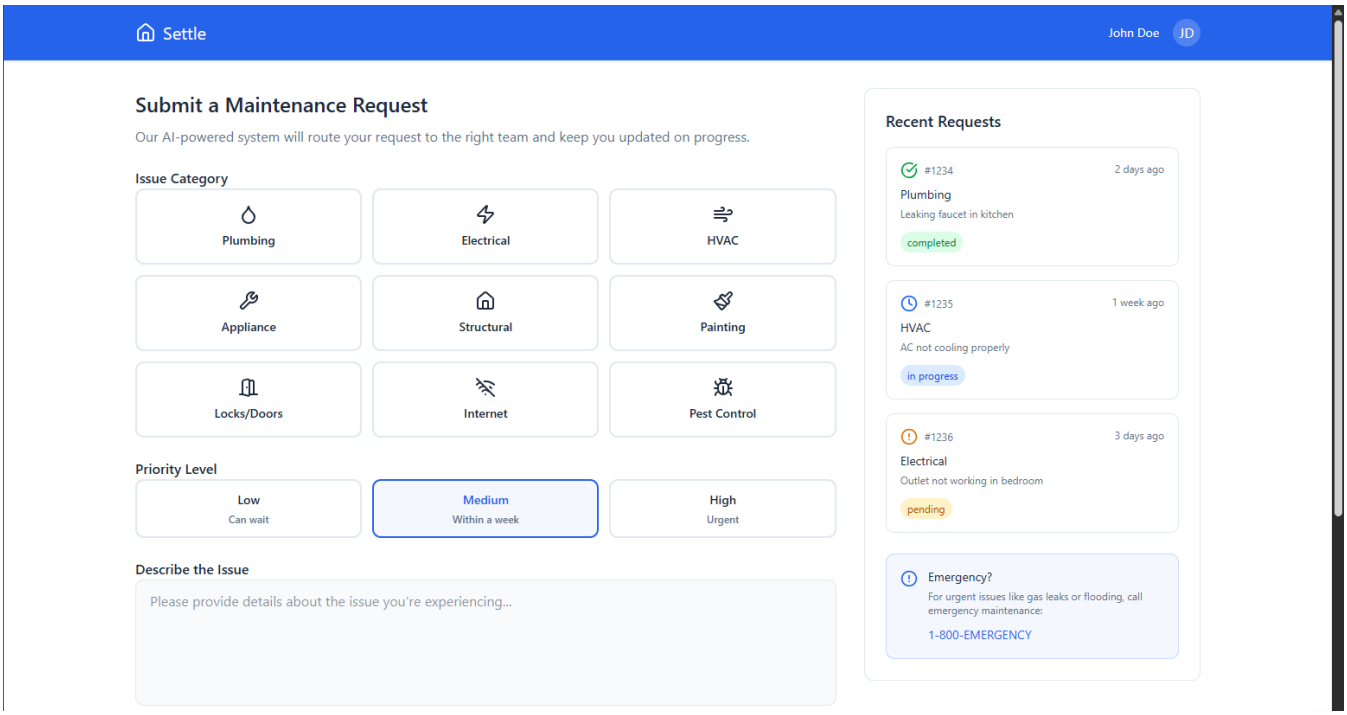

Figure 19 - Maintenance Request Page by Abhiram Annavajhula

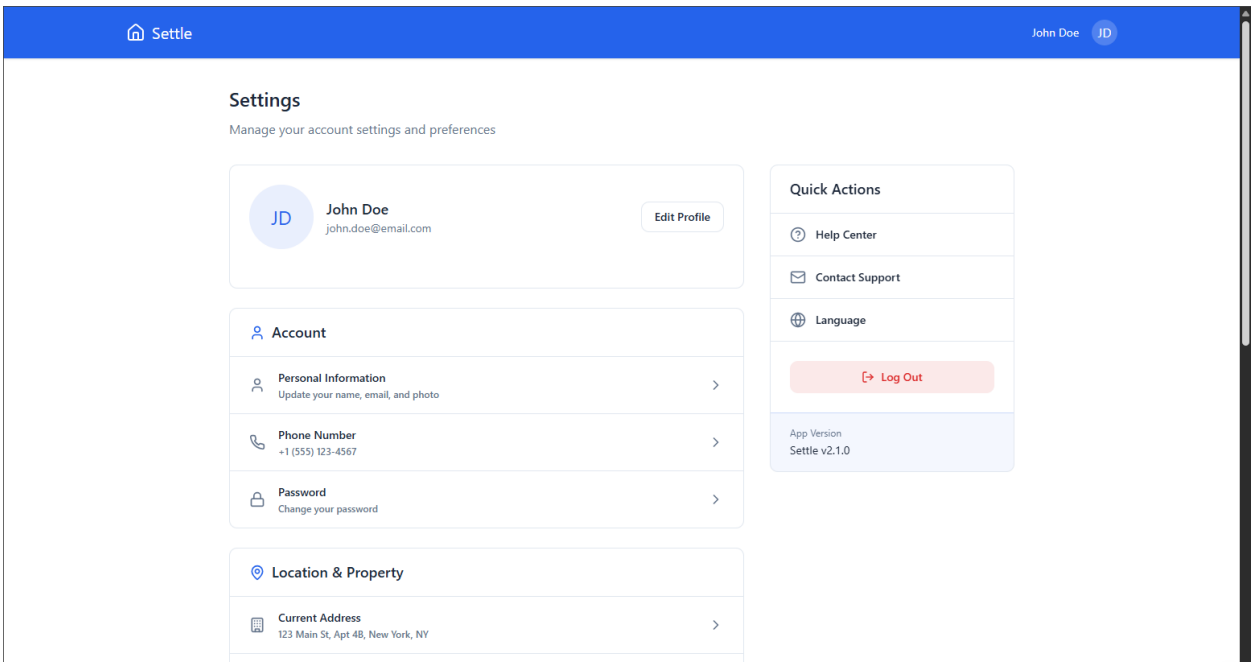

Figure 20 - Settings Page by Abhiram Annavajhula

4. Architecture Design

4.1 Introduction to Architecture Strategy

The overall architecture strategy of a software system is centered on organizing components in a structured and systematic way to support reliable development, validation, deployment, and long-term evolution. Settle will use a layered web application architecture which is designed to support personalized apartment recommendations, apartment search, listing details, and user account features. The system will be organized around major components such as those previously mentioned. This structure helps separate responsibilities across the system and makes it easier to develop, test, and maintain Settle.

For the development environment, the Settle team will build and update the frontend, backend services, recommendation features, and database logic in a controlled environment where these new features are implemented without affecting users. For the testing environment, the system will be validated using functional, integration, and performance testing to make sure features such as account creation, apartment search, recommendation generation, map views, and maintenance requests work efficiently together before the web application's release. For the production environment, Settle will run as the live version of the platform used by renters, where reliability, fast response time, security, and data accuracy are especially important. Therefore, this approach matches the software engineering idea that systems should support development, validation, and ongoing evolution in a structured manner.

4.2 Research Architecture Technologies and Methodologies

Software architecture focuses on defining the structure of a system and how its components interact. Architectural patterns like layered, client-server, and model-view-controller (MVC) help ensure scalability, performance, and maintainability. Technologies such as object-oriented programming, UML modeling, and design patterns that support building efficient and reusable systems.

Software development methodologies guide how systems are created and delivered. It explains plan-driven approaches like the waterfall model and agile methods like extreme programming, each suited for several types of projects. Combined with practices like requirements engineering and testing. These methodologies help ensure high-quality and adaptable software systems.

4.3 Context Diagram

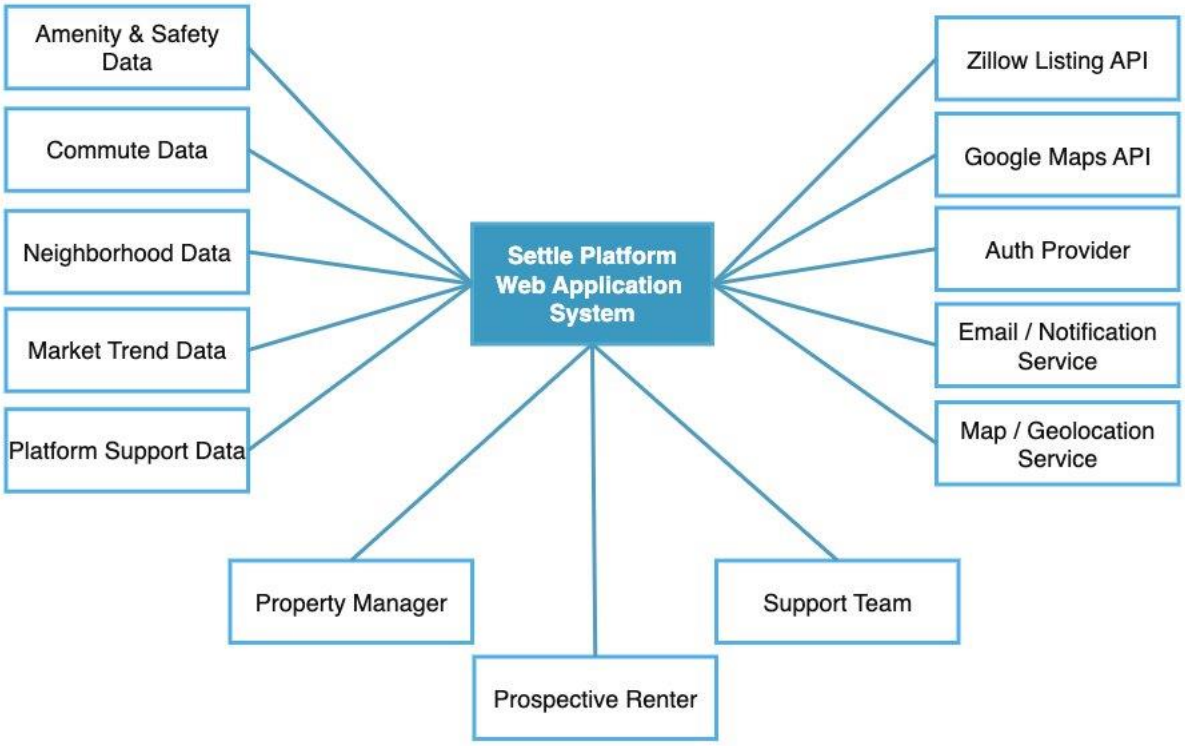

Figure 21 – Settle Web Application System Context Diagram by Jose Rogel & Rithanya Dhivakar

Figure 21 portrays the Settle platform system at the center, with lines that demonstrate the system's interaction with external data sources, APIs, and users. It highlights how the system gathers data such as safety, commute, and listings, to integrate with services like Zillow, maps, and auth, to serve different users like renters, property managers, and support teams.

4.4 Systems Architecture (Logical)

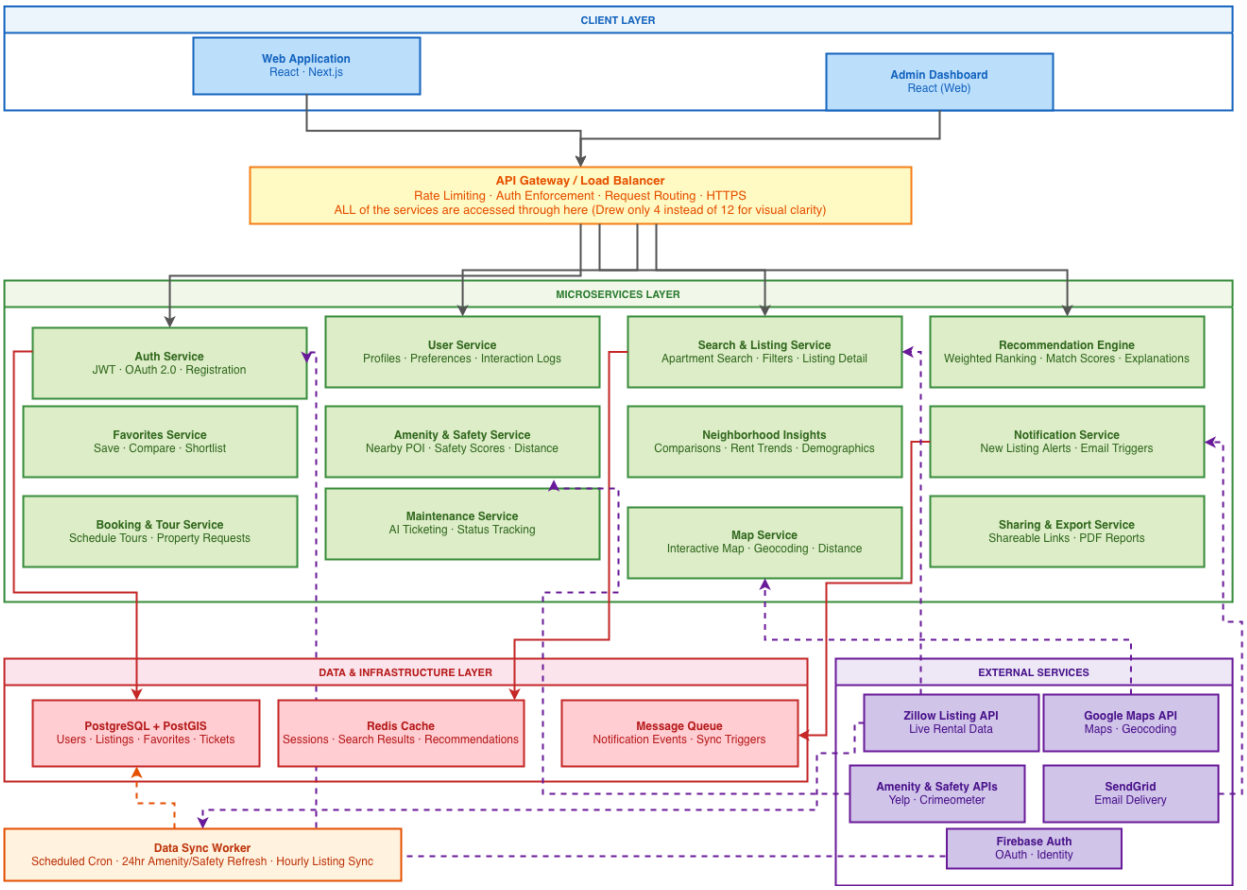

Figure 22 – Settle Web Application Logical Architecture Diagram by Abhiram Annavajhula & Jose Rogel

Figure 22 illustrates Settle's layered structure with the web app sending requests through an API gateway to microservices such as search, recommendations, users, and others, which then connects to the data layer and external APIs. Thus, highlighting a scalable, modular system design.

4.5 Systems Architecture (Physical)

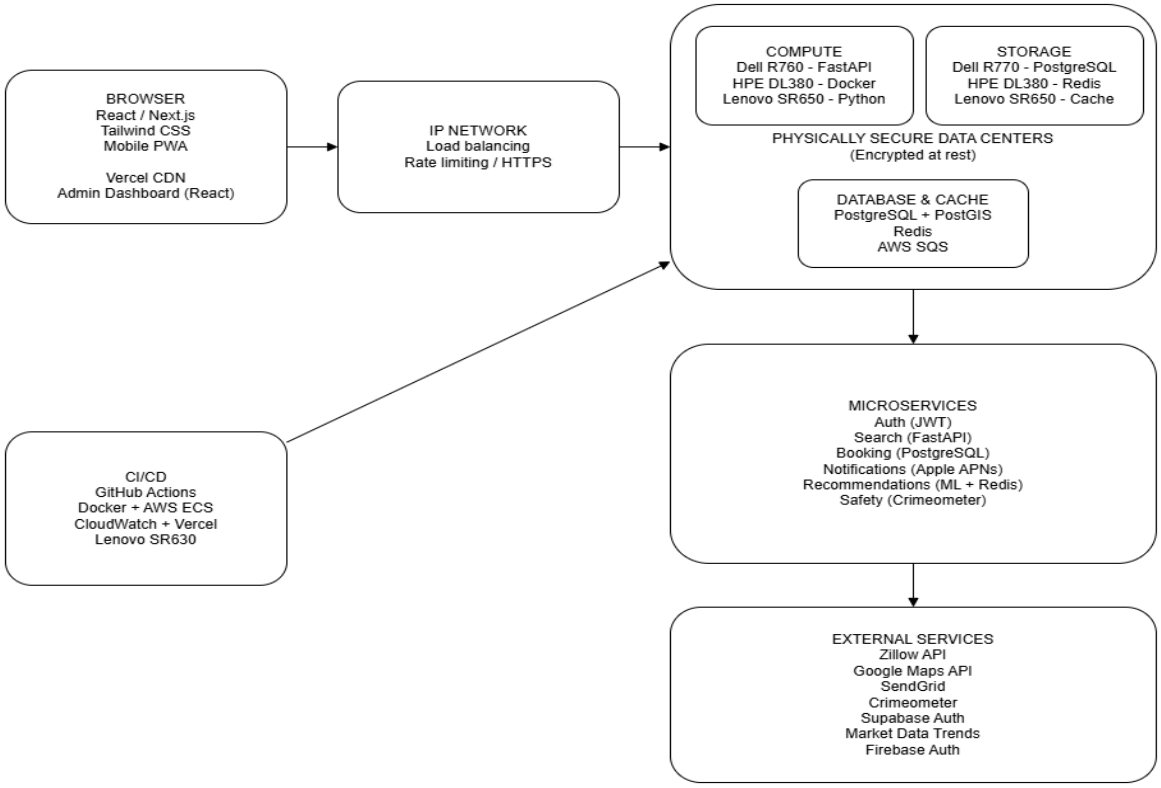

Figure 23 – Settle Web Application Physical Architecture Diagram by Shreya Sil & Abhiram Annavajhula

Figure 23 shows the physical architecture of the Settle web app on AWS, where users interact through the web app and admin dashboard via an API Gateway. The system runs containerized microservices in AWS ECS, with data stored in PostgreSQL, Redis, and AWS storage services. It also includes a CI/CD pipeline for deployment and integrates external APIs for listing, maps, and notifications.

4.6 Tech Stack

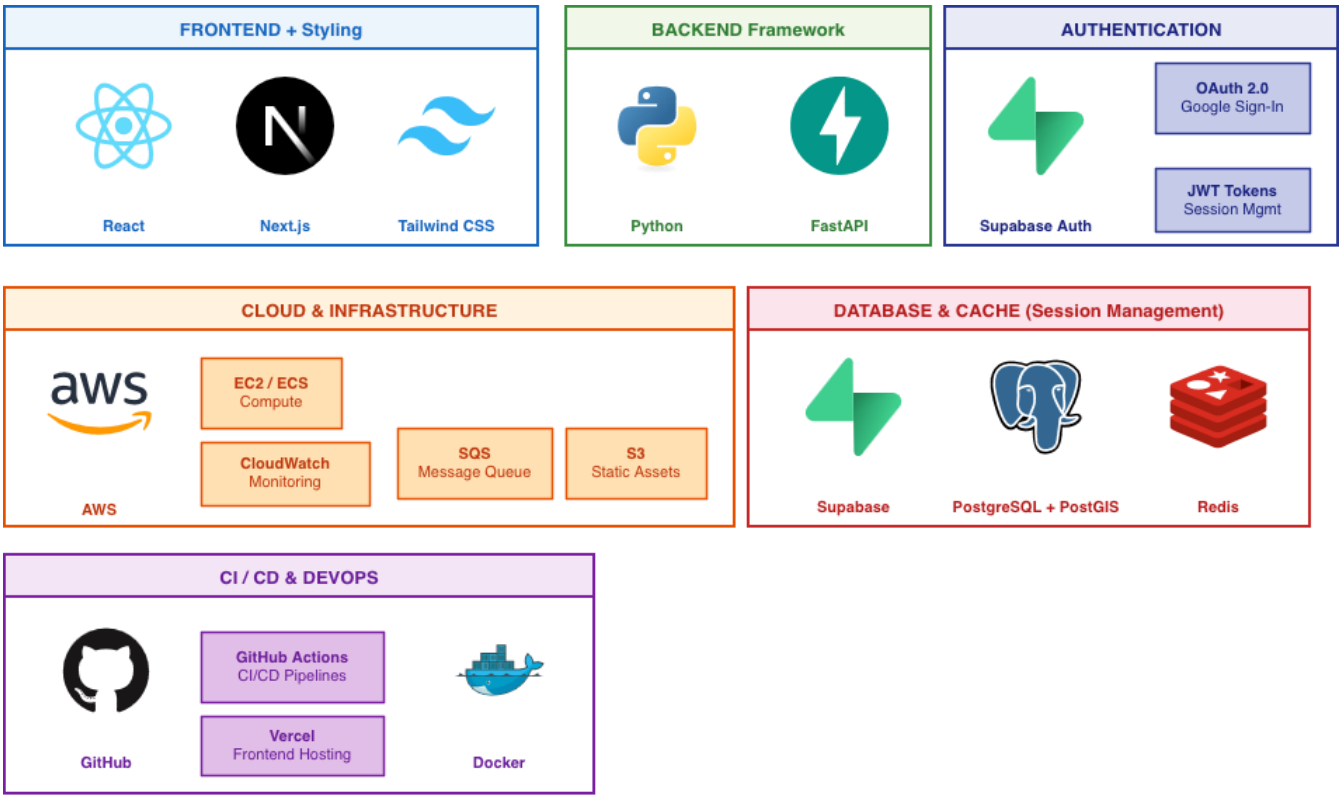

Figure 24 – Settle Web Application Technology Stack

Frontend (React Next.js, Tailwind CSS)
Supports fast, responsive UI for features like preference input, listing cards, and map views while meeting performance (≤3s load) and usability requirements.

Backend (Python, FastAPI)
Handles recommendation logic, filtering, and real-time updates efficiently, which is critical for generating results quickly and adapting to user interactions.

Authentication (Supabase Auth, OAuth 2.0, JWT)
Enables secure account creation, login, and session handling while protecting user data.

Cloud & Infrastructure (AWS: EC2/ECS, S3, SQS, CloudWatch)
Ensures scalability, reliability, and monitoring to support high user traffic and uptime requirements.

Database & Cache (PostgreSQL + PostGIS, Redis, Supabase)
Stores structured data like listings and user preferences, supports location-based queries for maps, and uses caching for faster response times.

CI / CD & DevOps (GitHub, GitHub Actions, Vercel, Docker)

Enables continuous updates, testing, and deployment without downtime, supporting system evolution and reliability.

4.7 UML Classes & Associations

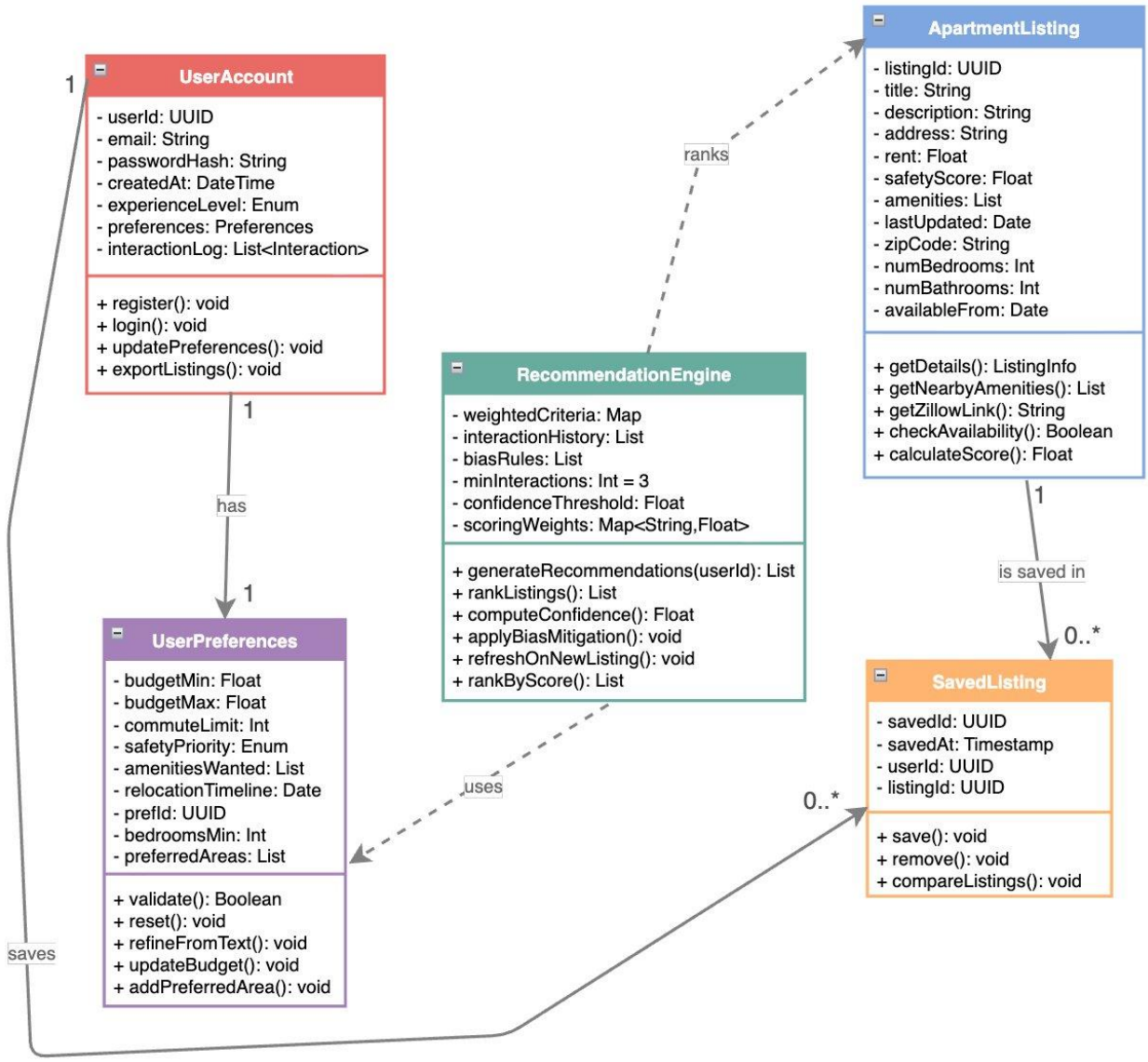

Figure 25 – Settle Web Application UML Classes & Associations Diagram by Rithanya Dhivakar & Shreya Sil

Figure 25 represents the core structure of the Settle web application, as it focuses on how the five main classes interact with each other. UserAccount connects to UserPreferences and SavedListing, while the RecommendationEngine uses this data to generate and rank ApartmentListings results. Therefore, this UML diagram highlights how user input is used to produce personalized recommendations.

5. Implementation

Part for Project Assignment 4

6. Software Testing

6.1 Research Testing Technologies and Methodologies

Testing Technologies and Methodologies focus on web functionality, APIs, performance, and usability, because the system includes a frontend interface, backend services, external data sources, and recommended features

Here are a few testing methodologies when it comes to trying out Settle:

Functional Testing: This ensures that the system features work according to the requirements, as described in this document. For Settle, specifically, that includes testing account creation, viewing the interactive map, scheduling apartment tours, submitting maintenance requests, and viewing neighborhood insights.

Integration Testing: This verifies that different components of the system work together correctly. That includes testing communication between frontend interfaces, backend server, recommendation engine, and external data sources that provide apartment listings and neighborhood data.

Performance Testing: This evaluates how the system performs under heavy load, which will very that recommendation results, map views, and listing searches load quickly without degrading performance

Here are also some testing technologies that can be used to test the application out as well:

Selenium: automated browser testing verifies features such as login, apartment search, and viewing maps work correctly across different browsers

Postman: API Testing verify communication between the frontend, backend services, and external housing data sources

Apache JMeter: Performance testing à simulates thousands of users accessing the platform simultaneously à ensures the system meets scalability requirements

Cypress: End-to-end testing simulates real user interactions, such as creating an account, searching for apartments, and scheduling tours.

6.2 Introduction on Testing Plan

Testing is an important part of software engineering because it helps make sure the system works correctly and meets the requirements set during development. In the Settle platform, testing will

be done to check the key features, such as searching for listings, viewing property details, contacting property owners, and saving favorite listings, work properly. This section will cover how the system will be tested for functionality, usability, and overall performance. By testing the system carefully, the team can find and fix problems before the platform is released to users.

6.3 Functional & Non-Functional Test Cases for Requirements

Rithanya Dhivakar

TC-01

Description: Verifies that the system allows users to enter housing preferences using structured inputs.

1. Open the User Preferences page.
2. Adjust the budget slider.
3. Select commute tolerance.
4. Select safety priority.
5. Choose amenities using checkboxes.
6. Select renter experience level.
7. Submit preferences.

Expected result: The system accepts all inputs and successfully records the user's preferences.

TC-02

Description: Verify that the system generates recommendation results based on user preferences.

1. Enter preferences on the preferences page.
2. Submit preferences.
3. Navigate to the results page.

Expected Result: The system displays recommendation listing cards showing price, location, commute estimate, safety rating, and a match indicator aligned with the user's preferences.

TC-03

Description: Verify that the system provides clear explanations for each recommendation.

1. Open the recommendation results page.
2. Select a recommended listing.

Expected Result: The listing displays a concise explanation highlighting the primary factors influencing the recommendation and any trade-offs.

TC-04

Description: Verify that users can view additional listing information.

1. Open the recommendation results page.
2. Click the "View Details" option on a listing.

Expected Result: The system displays additional listing information such as neighborhood context or detailed listing information.

TC-05
Description: Verify that users can refine recommendation focus through free-text input.

1. Open the dashboard or results page.
2. Enter a refinement request in the free-text input (example: “closer to downtown”).
3. Submit the request.

Expected Result: The system interprets the input, updates relevant preferences, and regenerates recommendation results.

TC-06
Description: Verify that recommendation results load within the required response time.

1. Submit user preferences.
2. Measure the time taken for the results page to load.

Expected Result: Recommendation results appear within 3 seconds under normal operating conditions.

TC-07
Description: Verify that updated recommendations are generated quickly after a preference change.

1. Open the recommendation results page.
2. Modify a preference or submit a refinement input.
3. Measure the time for updated results to appear.

Expected Result: Updated recommendation results appear within 2 seconds after the change is submitted.

Jose Rogel
TC-08
Description: Verify that searching by zip code returns a filtered list of matching apartment listings.

1. Navigate to the Settle search page.
2. Enter "75080" into the search bar.
3. Submit the search query.Expected Result: A filtered list of apartment listings located within zip code 75080 is displayed.

TC-09
Description: Verify that applying filters correctly narrows apartment search results.

1. Navigate to the search results page.
2. Set the Total Monthly Cost filter to a maximum of $1,500.
3. Set the Bedrooms filter to 1 bedroom.
4. Enable the Pet Policy filter.
5. Submit the filtered search.

Expected Result: Only listings that satisfy all three active filters are displayed in the results.

TC-10
Description: Verify that a listing detail page displays all required information.

1. Navigate to the search results page.

2. Click "View Details" on any listing card.

Expected Result: The listing detail page displays rent, address, available amenities, neighborhood safety score, a direct message option, and secondary contact information for the property.

TC-11

Description: Verify that nearby amenities are displayed with correct distance labels for a selected listing.

1. Open any apartment listing detail page.
2. Scroll to the Nearby Amenities section.

Expected Result: At least three nearby amenity categories are displayed, each with a distance label in miles from the listing.

TC-12

Description: Verify that the Contact Us form successfully submits a user inquiry.

1. Navigate to the Contact Us page.
2. Fill in the Full Name field.
3. Fill in the Email Address field.
4. Select a subject from the dropdown.
5. Enter a message in the Message field.
6. Click the Submit button.

Expected Result: A success confirmation message is displayed and the inquiry is routed to the support team via email.

TC-13

Description: Verify that apartment search results load within the required response time.

1. Navigate to the search page.
2. Enter a search query with at least three active filters applied.
3. Submit the search and begin timing.

Expected Result: Search results are fully loaded and displayed within 2.5 seconds under normal network conditions.

TC-14

Description: Verify that the apartment search and listing detail pages meet WCAG 2.1 AA accessibility standards.

1. Open the apartment search results page.
2. Run an automated accessibility audit using a WCAG 2.1 AA compliance tool.
3. Repeat the audit on the apartment listing detail page.

Expected Result: Both pages return zero critical or serious accessibility violations against WCAG 2.1 AA criteria.

Abhiram Annavajhula

TC-15

Description: Verify that a user can create a personal account using email registration.

Steps:

1. Navigate to the registration page.
2. Enter a valid email address and password.
3. Submit the registration form.

Expected Result:
The system creates a new account and confirms successful registration.

TC-16
Description: Verify that recommended apartments appear on the interactive map.
Steps:
1. Log into the system.
2. Navigate to the recommendations page.
3. Select the map view option.

Expected Result:
The system displays apartments and neighborhoods on an interactive map.

TC-17
Description: Verify that a user can schedule a tour using the request form.
Steps:
1. Select an apartment listing.
2. Click "Schedule Tour."
3. Fill out the request form with date and contact details.
4. Submit the form.

Expected Result:
The system sends the request and confirms that the tour request was submitted.

TC-18
Description: Verify that users can track maintenance request status.
Steps:
1. Log into the system.
2. Navigate the maintenance request section.
3. Select a previously submitted request.

Expected Result:
The system displays the request status (Submitted, In Progress, or Resolved).

TC-19
Description: Verify that neighborhood trend data is displayed.
Steps:
1. Search for a neighborhood.
2. Open the neighborhood insights section.
3. View trend information.

Expected Result:
The system displays average rent trends and demographic data over time.

TC-20

Description: Verify the system supports 10,000 concurrent users.
Steps:
1. Simulate 10,000 users accessing the system simultaneously.
2. Perform actions such as searching or loading listings.

Expected Result:
The system remains functional with no significant performance degradation.

TC-21
Requirement: NFR-03
Description: Verify that new users can complete preference setup within 5 minutes.
Steps:
1. Create a new account.
2. Start the preference setup process.
3. Complete all required preference fields.

Expected Result:
The user can complete the setup process within 5 minutes without assistance.

Shreya Sil
TC-22
Requirement: FR-01
Description: Verify that users can search for apartments by zip code, neighborhood name, or keyword.
Steps:
1. Open the apartment search page.
2. Enter a zip code, neighborhood name, or keyword in the search bar.
3. Click the search button.

Expected Result:
The system displays a filtered list of apartment listings that match the entered search input.
TC-23
Requirement: FR-02
Description: Verify that users can filter apartment search results by total monthly cost, number of bedrooms, pet policy, and proximity to workplace.
Steps:
1. Search for apartments.
2. Apply filters for monthly cost, number of bedrooms, pet policy, and proximity to workplace.
3. View the filtered results.

Expected Result:
The system displays apartment listings that match the selected filter criteria.
TC-24
Requirement: FR-03
Description: Verify that the apartment listing page displays detailed property information.
Steps:
1. Search for apartments.

2. Select an apartment listing.
3. Open the listing details page.

Expected Result:
The system displays rent, address, amenities, neighborhood safety score, direct messaging option, and secondary contact information.

TC-25
Requirement: FR-04
Description: Verify that nearby amenities are displayed for a selected apartment.
Steps:

1. Open an apartment listing.
2. Navigate to the nearby amenities section.
3. View the listed amenities.

Expected Result:
The system displays nearby grocery stores, gyms, parks, restaurants, coworking spaces, and entertainment venues along with their distance from the apartment.

TC-26
Requirement: FR-05
Description: Verify that users can save apartment listings to a persistent favorites list accessible from their profile.
Steps:

1. Log into the system.
2. Open an apartment listing.
3. Click the save to favorites option.

Expected Result:
The apartment listing is saved to the user's favorites list and remains accessible from the user profile.

TC-27
Requirement: NFR-01
Description: Verify that updated recommendations are generated within 3 seconds of user input submission.
Steps:

1. Enter or update user preferences.
2. Submit the input.
3. Measure the response time.

Expected Result:
The system generates and displays updated recommendations within 3 seconds.

TC-28
Requirement: NFR-02
Description: Verify that all user data is encrypted both in transit and at rest.
Steps:

1. Log into the system.
2. Submit user data through the platform.
3. Inspect the system's security configuration.

Expected Result:
User data is encrypted during transmission and while stored in the system.

This version matches your teammates way better.

6.4 Requirements Traceability Matrix

Requirement	Requirement Description	Test Cases	Test Case Description	Owner
FR-01	The system shall collect user preferences using structured inputs, including sliders, checkboxes, and selection controls, to capture constraints such as budget, commute tolerance, safety priority, amenities, and renter experience level.	TC-01	Verifies that the system allows users to enter housing preferences using structured inputs.	Rithanya Dhivakar
FR-02	The system shall generate and display recommendation results that reflect the user's stated preferences and strongest constraints, including budget range, commute tolerance, and safety priority, and present each recommendation as a listing card showing key details such as price, location, commute, safety rating, and a match indicator.	TC-02	Verify that the system generates recommendation results based on user preferences.	Rithanya Dhivakar
FR-04	The system shall display plain-language explanations for each recommendation that describe the primary factors influencing the result and any notable trade-offs, and shall present these explanations as concise bullet-style highlights summarizing the primary influencing factors.	TC-03	Verify that the system provides clear explanations for each recommendation.	Rithanya Dhivakar
FR-05	The system shall allow users to view additional details behind a	TC-04	Verify that users can view	Rithanya Dhivakar

	recommendation, such as neighborhood context or listing information, without requiring them to view raw or complex data.		additional listing information.	
FR-09	The system shall allow users to refine or change recommendation focus through a free-text input, and shall interpret the input to update relevant preferences and regenerate recommendation results.	TC-05	Verify that users can refine recommendation focus through free-text input.	Rithanya Dhivakar
NFR-01	The system shall load and display recommendation results within 3 seconds after the results page is requested under normal operating conditions.	TC-06	Verify that recommendation results of load within the required response time.	Rithanya Dhivakar
NFR-02	The system shall regenerate and display updated recommendation results within 2 seconds after a user modifies preferences or submits a preference of refinement.	TC-07	Verify that updated recommendations are generated quickly after a preference to change.	Rithanya Dhivakar
FR-11	The system must allow users to search for apartments by zip code, neighborhood name, or keyword and return a filtered list of matching listings.	TC-08	Verify that searching by zip code returns a filtered list of matching apartment listings.	Jose Rogel
FR-12	The system must allow users to filter apartment search results by total monthly cost, number of bedrooms, pet policy, and proximity to workplace.	TC-09	Verify that applying filters correctly narrows apartment search results.	Jose Rogel
FR-13	The system must display a detailed apartment listing page showing rent, address, available amenities, neighborhood safety score, a direct messaging	TC-10	Verify that a listing detail page displays all required information.	Jose Rogel

	option, and secondary contact information for the property.			
FR-14	The system must display nearby amenities for a selected apartment, including grocery stores, gyms, parks, restaurants, coworking spaces, and entertainment venues, along with their distance from the listing.	TC-11	Verify that nearby amenities are displayed with correct distance labels for a selected listing.	Jose Rogel
FR-17	The system must provide a Contact Us page with a form that allows users to submit inquiries, which are then routed to the platform support team via email.	TC-12	Verify that the Contact Us form successfully submits a user inquiry.	Jose Rogel
NFR-05	The system must load apartment search results within 2.5 seconds of a user submitting a query under normal network conditions, regardless of the number of active filters applied.	TC-13	Verify that apartment search results load within the required response time.	Jose Rogel
NFR-06	The apartment search, listing, amenity, About, and Contact Us pages must meet WCAG 2.1 AA accessibility standards, ensuring usability for users with disabilities across all devices.	TC-14	Verify that the apartment search and listing detail pages meet WCAG 2.1 AA accessibility standards.	Jose Rogel
FR-31	The system shall allow users to create and manage a personal account using email registration and secure login authentication	TC-15	Verify that a user can create a personal account using email registration.	Abhiram Annavajhula
FR-33	The system shall provide an interactive map view that displays recommended apartments neighborhoods geographically	TC-16	Verify that recommended apartments appear on the interactive map.	Abhiram Annavajhula
FR-35	The system shall allow users to schedule apartment tours or request additional information	TC-17	Verify that a user can schedule a	Abhiram Annavajhula

	from property managers through an integrated request form		tour using the request form.	
FR-37	The system shall provide status tracking for submitted maintenance requests, allowing users to view updates such as submitted, in progress, or resolved	TC-18	Verify that users can track maintenance request status.	Abhiram Annavajhula
FR-39	The system shall allow users to view neighborhood trend insights, including average rent changes and population demographics over time	TC-19	Verify that neighborhood trend data is displayed.	Abhiram Annavajhula
NFR-09	The system shall support at least10,000 concurrent users without significant degradation in system performance	TC-20	Verify the system supports 10,000 concurrent users.	Abhiram Annavajhula
NFR-11	The system interface shall allow new users to complete the initial preference setup within 5 minutes without external assistance	TC-21	Verify that new users can complete preference setup within 5 minutes.	Abhiram Annavajhula
FR-01	The system shall allow users to search for apartments by zip code, neighborhood name, or keyword and return a filtered list of matching listings.	TC-22	Verify that users can search for apartments by zip code, neighborhood name, or keyword and receive matching listings.	Shreya Sil
FR-02	The system shall allow users to filter apartment search results by total monthly cost, number of bedrooms, pet policy, and proximity to workplace.	TC-23	Verify that users can filter apartment search results based on selected criteria.	Shreya Sil
FR-03	The system shall display a detailed apartment listing page showing rent, address, available amenities, neighborhood safety score, a direct messaging option, and secondary contact information for the property.	TC-24	Verify that the apartment listing page displays detailed property information.	Shreya Sil

FR-04	The system shall display nearby amenities for a selected apartment, including grocery stores, gyms, parks, restaurants, coworking spaces, and entertainment venues, along with their distance from the listing.	TC-25	Verify that nearby amenities and distances are displayed for a selected apartment.	Shreya Sil
FR-05	The system shall allow users to save apartment listings to a persistent favorites list accessible across devices from their profile.	TC-26	Verify that users can save apartment listings to a favorites list accessible from their profile.	Shreya Sil
NFR-01	The system shall generate updated recommendations within 3 seconds of user input submission.	TC-27	Verify that updated recommendations are generated within the required response time.	Shreya Sil
NFR-02	The system shall encrypt all user data both in transit and at rest.	TC-28	Verify that user data is encrypted during transmission and storage.	Shreya Sil

7. Software Evolution

Part for Project Assignment 4

8. Glossary of Terms

Term	Definition
Amenity Data	Information about nearby facilities such as grocery stores, gyms, parks, restaurants, coworking spaces, and entertainment venues associated with a listing.
Brand & Creative Lead	A stakeholder responsible for defining how the platform communicates its mission, branding, and user messaging across pages such as the About page.
Business Operations Lead	A stakeholder responsible for defining operational policies, recommendation rules, and how the system handles edge cases such as restrictive user preferences.
Compliance Lead	A stakeholder responsible for ensuring the system meets legal and regulatory requirements such as accessibility standards and fairness guidelines.
Contact Form	A feature that allows users to submit inquiries or support requests directly to the platform's support team.
Data Scientist	A stakeholder responsible for designing and evaluating the recommendation algorithm, determining which user preferences influence results, and ensuring fairness in recommendations.
Favorites List	A persistent list where users can save apartment listings for later review or comparison.
Functional Requirement (FR)	A requirement that defines a specific behavior, feature, or capability that the system must perform.
Interaction History	A record of user actions such as viewed listings, saved listings, and refinements that can influence future recommendations.
Listing Card	A visual summary of a rental listing displayed on the results page, showing key details such as price, location, commute estimate, safety rating, and match indicator.
Match Indicator	A metric representing how closely a listing aligns with a user's stated preferences and constraints.
Neighborhood Safety Score	A rating or metric representing the relative safety level of the neighborhood surrounding a property listing.
Non-Functional Requirement (NFR)	A requirement that defines system qualities such as performance, reliability, usability, security, or scalability.
Preference Refinement	A feature that allows users to modify recommendation constraints through structured inputs or free-text input.
Recommendation Engine	The system component that analyzes user preferences and interaction history to generate ranked housing recommendations.

Recommendation Explanation	A plain-language description explaining the primary factors influencing why a listing was recommended.
Recommendation Personalization	The process of adapting recommendation results based on user behavior and interaction patterns.
Requirements Traceability Matrix (RTM)	A table that maps system requirements to corresponding test cases to ensure each requirement is validated during testing.
Sequence Diagram	A UML diagram that illustrates the order of interactions between system components over time as a process is executed.
Settle Platform	The web application is designed to help relocate renters find apartment recommendations based on preferences, constraints, and behavioral interactions.
Site Map	A diagram showing the structure of the application and how users navigate between pages and features.
Use Case Diagram	A UML diagram that illustrates the interactions between users (actors) and the system's functions.
User Preference	Inputs provided by the user, such as budget range, commute tolerance, safety priority, and amenities, are used to personalize recommendation results.